



WELCOME TO THE PYTHON DECAL!

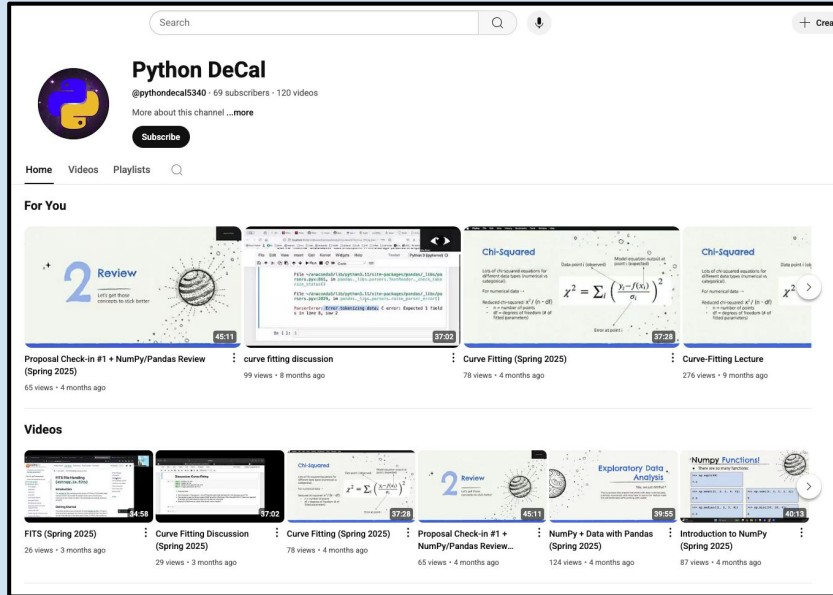
Discussion: GitHub
[Class 4] September 10th, 2025



LECTURES WILL BE RECORDED

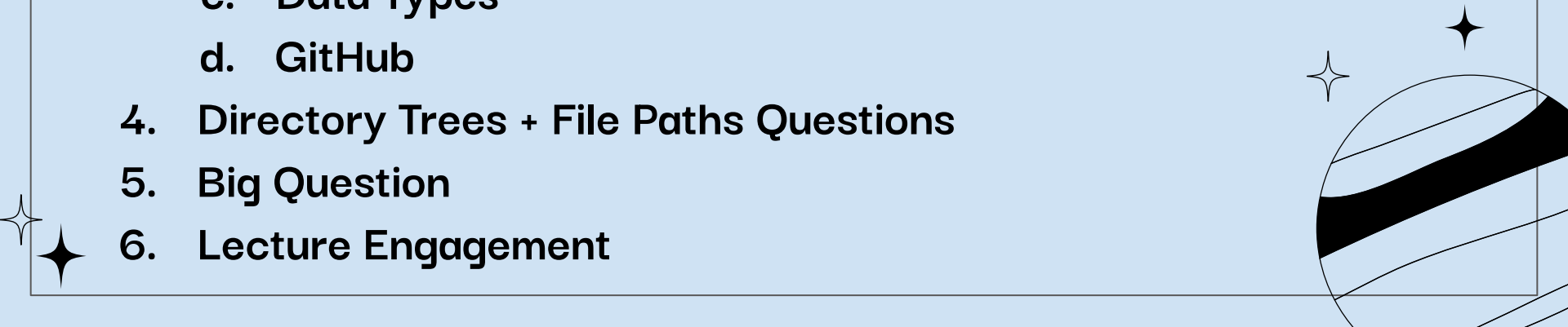
This class will be
recorded and posted to
our YouTube Channel!

Subscribe :)





overview FOR TODAY!

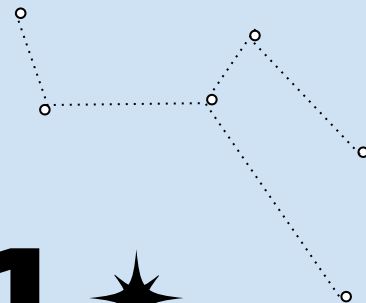
1. Announcements
 2. Warm-Up
 3. Lecture Review
 - a. VS Code
 - b. Terminal + Command Line
 - c. Data Types
 - d. GitHub
 4. Directory Trees + File Paths Questions
 5. Big Question
 6. Lecture Engagement
- 

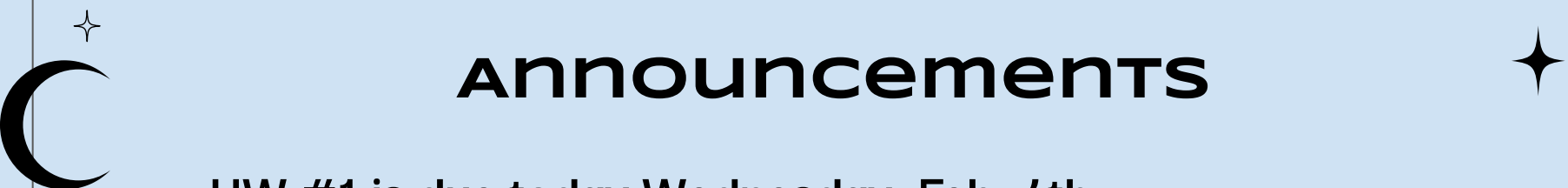


★ O1 ★

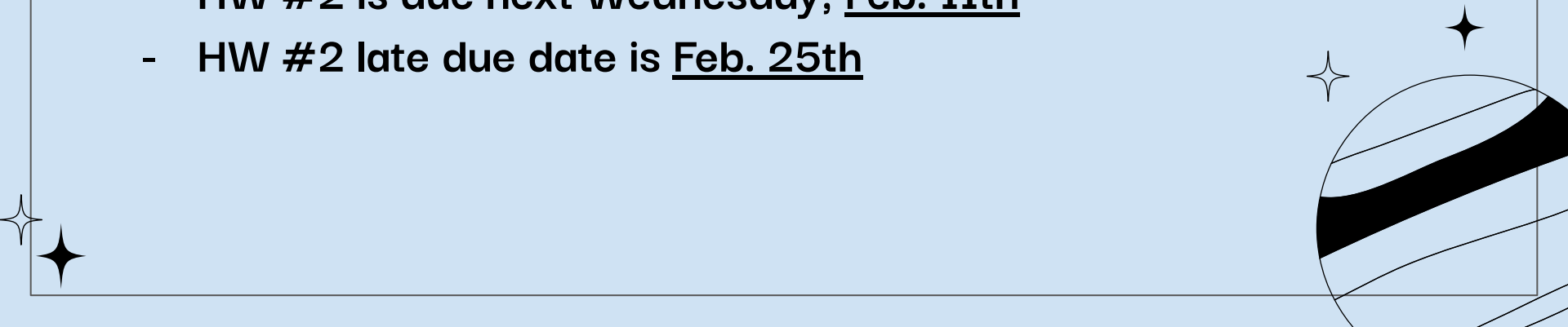
Announcements

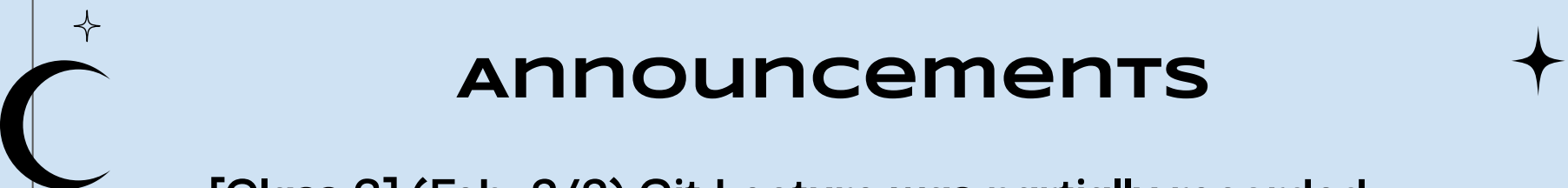
:)) :)) :)) :)) :)) :)) :)) :)) :)) :))





Announcements

- HW #1 is due today Wednesday, Feb. 4th
 - HW #1 late due date is Feb. 18th
 - Next, next Wednesday
 - HW #2 was released on Monday, Feb. 2nd
 - HW #2 is due next Wednesday, Feb. 11th
 - HW #2 late due date is Feb. 25th
- 



Announcements

- [Class 3] (Feb. 2/2) Git Lecture was partially recorded
 - Video not uploaded yet
 - Can record a video over the weekend if there's a large desire for it
 - Slides are very detailed, please read through them if you missed the lecture!
- 



★ 02 ★

Warm-Up

Let's do a little review





warm-UP questions



Please answer these on a piece of paper:

- 1) What three commands allow you to save your code?
- 2) What does the command git status do?
- 3) Would these evaluate to True or False? Why?
 - `x = 6, print(x > 2 or x < 5)`
 - `print(bool([]))`
 - `print(True and False)`
- 4) List all the datatypes you know with an example for each.



-





warm-up questions Answers



1) `git add .`
`git commit -m "message"`
`git push origin main`

2) shows the current state
of your working directory
and staging area

3) True
False
False

4) `integer = 3`
`float = 3.5`
`list = [1, 2, 3]`
`dict = {"num": 4}`
`tuple = (1, 2)`

...

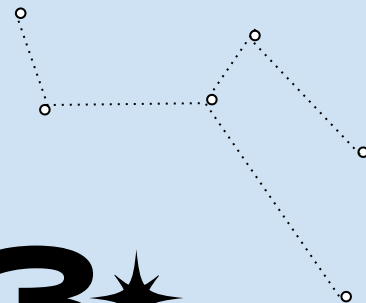




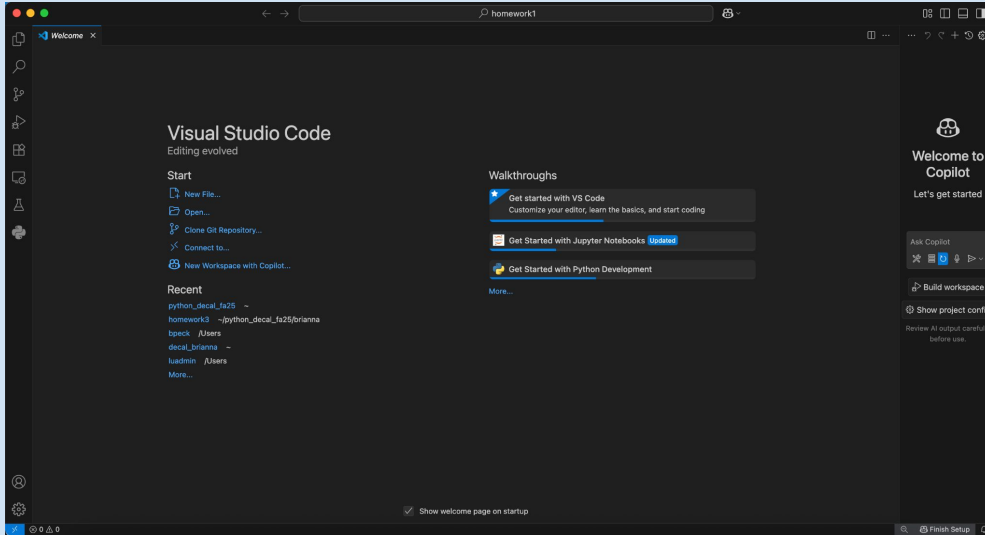
★ 03 ★

Lecture Review

Command Line + GitHub



REVIEW: VS CODE

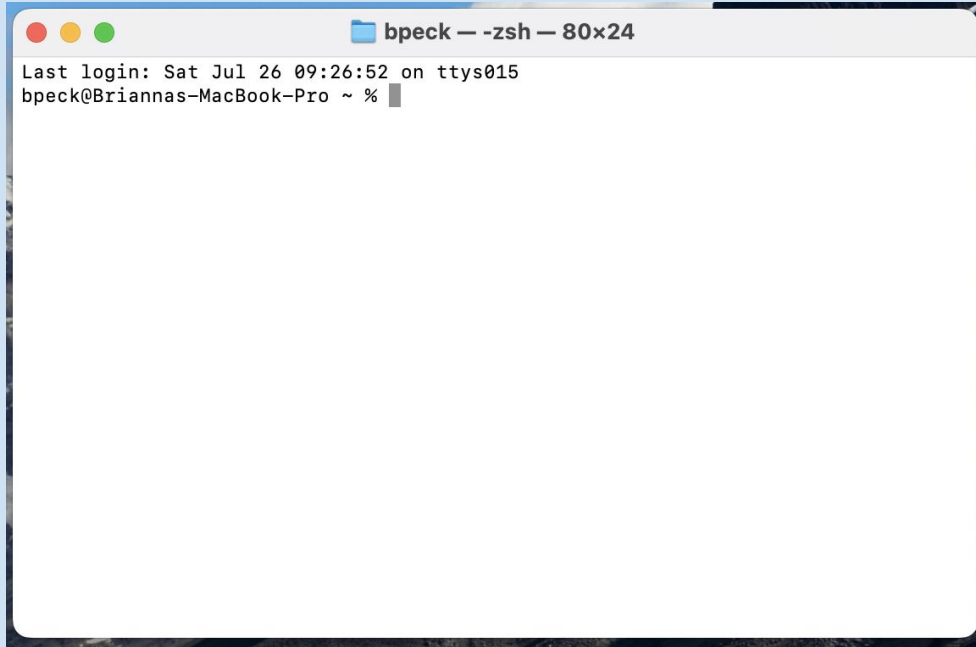


VS Code is a very common, open-source, code-editing platform

You will use it to work on homeworks and eventually Final Projects!

Refer [Class 2] slides for how to use VS Code

REVIEW: Terminal

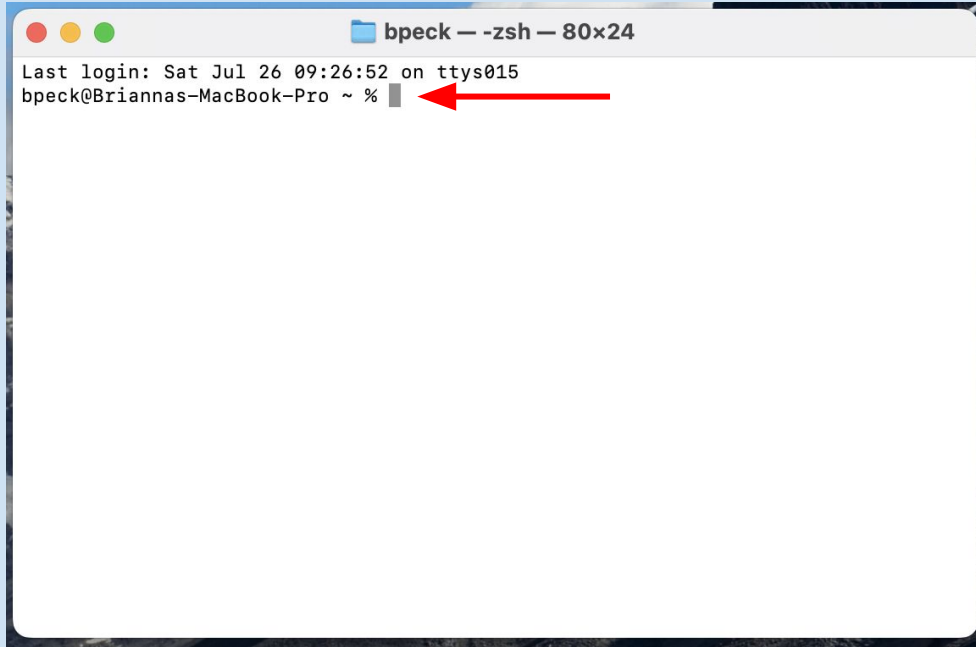


The terminal is a way to more directly interact with your computer

Use the terminal to save your work and run your code.

Refer [Class 2, 3] slides for how to use the terminal

REVIEW: command Line

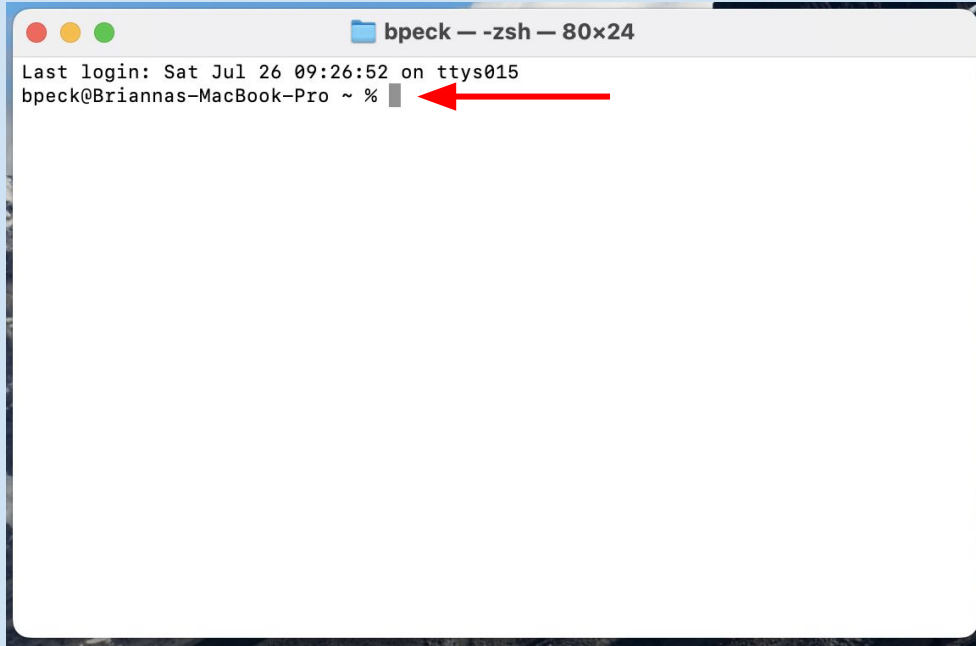
A screenshot of a macOS terminal window titled "bpeck — -zsh — 80x24". The window shows the output of a login session: "Last login: Sat Jul 26 09:26:52 on ttys015" followed by the prompt "bpeck@Briannas-MacBook-Pro ~ %". A red arrow points to the prompt character "%".

```
bpeck — -zsh — 80x24
Last login: Sat Jul 26 09:26:52 on ttys015
bpeck@Briannas-MacBook-Pro ~ %
```

The command line is the spot on the terminal where you input commands

The percentage % symbol means your command line is awaiting its next prompt

REVIEW: command Line

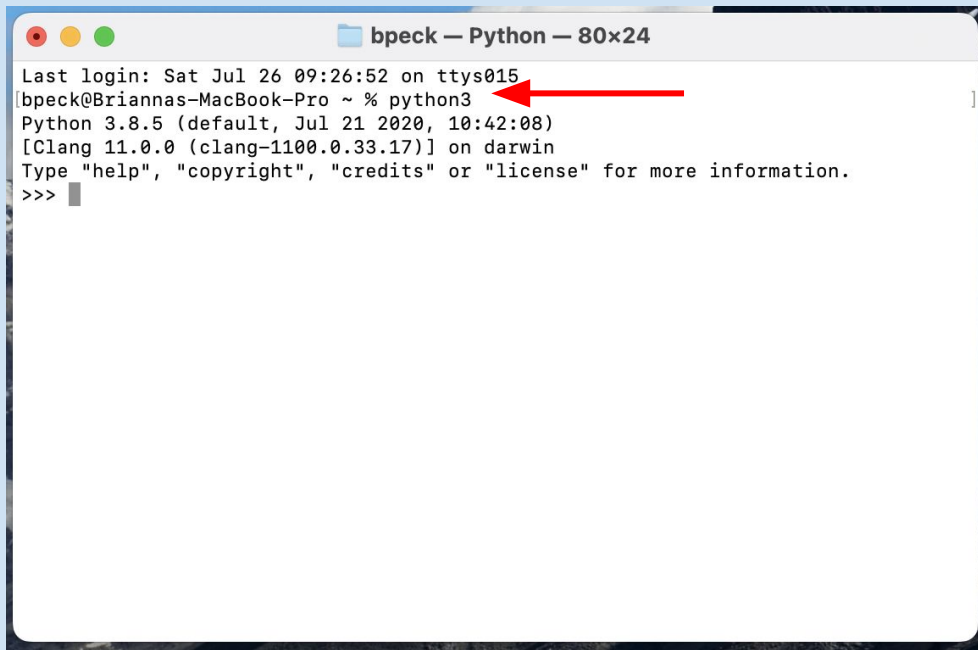
A screenshot of a macOS terminal window titled "bpeck — -zsh — 80x24". The window shows the login message "Last login: Sat Jul 26 09:26:52 on ttys015" and the prompt "bpeck@Briannas-MacBook-Pro ~ %". A red arrow points to the prompt character "%".

```
bpeck — -zsh — 80x24
Last login: Sat Jul 26 09:26:52 on ttys015
bpeck@Briannas-MacBook-Pro ~ %
```

Common Commands:

- pwd
- ls
- cd
- mkdir
- mv
- cat
- touch
- nano
- rm

REVIEW: command Line



```
bpeck — Python — 80x24
Last login: Sat Jul 26 09:26:52 on ttys015
bpeck@Briannas-MacBook-Pro ~ % python3
Python 3.8.5 (default, Jul 21 2020, 10:42:08)
[Clang 11.0.0 (clang-1100.0.33.17)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

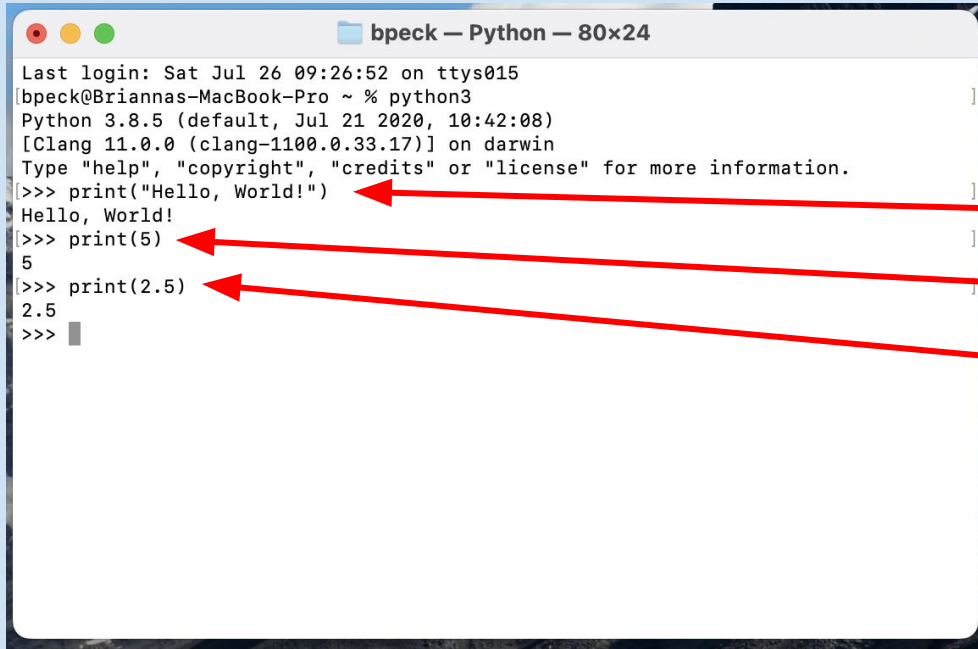
To run Python with your command line, you have two main options:

Call: python3 <script>

Call: python 3

- Which puts you in a python-speaking terminal

REVIEW: command Line



```
bpeck — Python — 80x24
Last login: Sat Jul 26 09:26:52 on ttys015
bpeck@Briannas-MacBook-Pro ~ % python3
Python 3.8.5 (default, Jul 21 2020, 10:42:08)
[Clang 11.0.0 (clang-1100.0.33.17)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
[>>> print("Hello, World!")
Hello, World!
[>>> print(5)
5
[>>> print(2.5)
2.5
[>>> ]
```

In class, we have covered three Python data types so far:

- string = text
- integer = whole number
- float = fractional #

You will explore more data types, expressions, operations in the homework

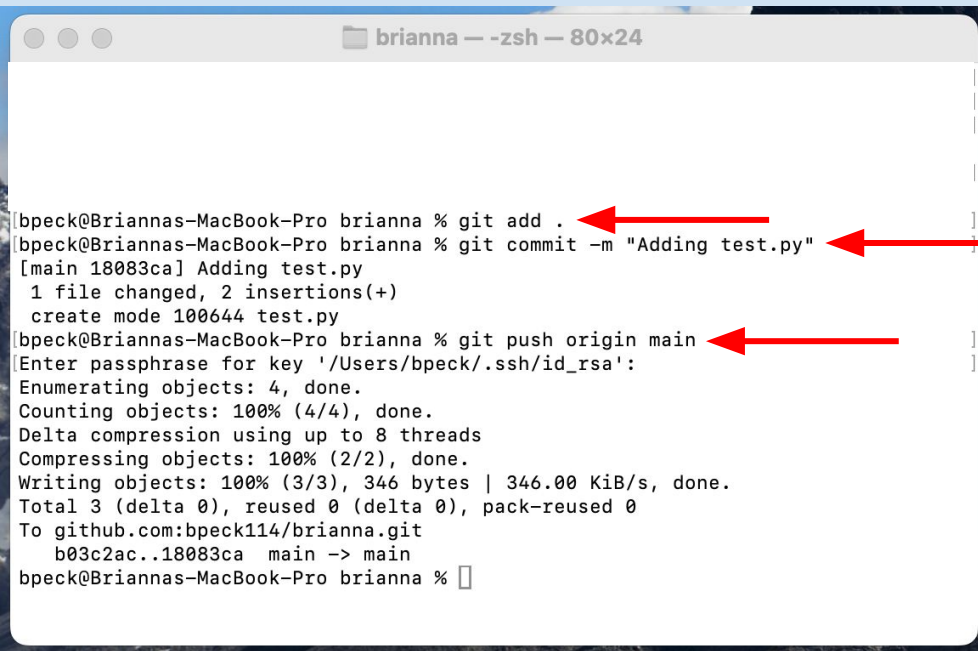
REVIEW: GITHUB

Last class we created two repositories:

- Local = on your comp.
- Remote = on internet

To save your local change:

- `git add .`
- `git commit -m "message"`
- `git push origin main`



```
brianna — -zsh — 80x24

bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Adding test.py"
[main 18083ca] Adding test.py
 1 file changed, 2 insertions(+)
 create mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 346 bytes | 346.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:bpeck114/brianna.git
 b03c2ac..18083ca main -> main
bpeck@Briannas-MacBook-Pro brianna %
```

Red arrows point to the following lines in the terminal output:

- `git add .`
- `git commit -m "Adding test.py"`
- `git push origin main`

REVIEW: GITHUB (LOCAL)

Last class we created two repositories:

- Local = on your comp.
- Remote = on internet

To save your local change:

- git add .
- git commit -m "message"
- git push origin main

```
brianna — -zsh — 80x24

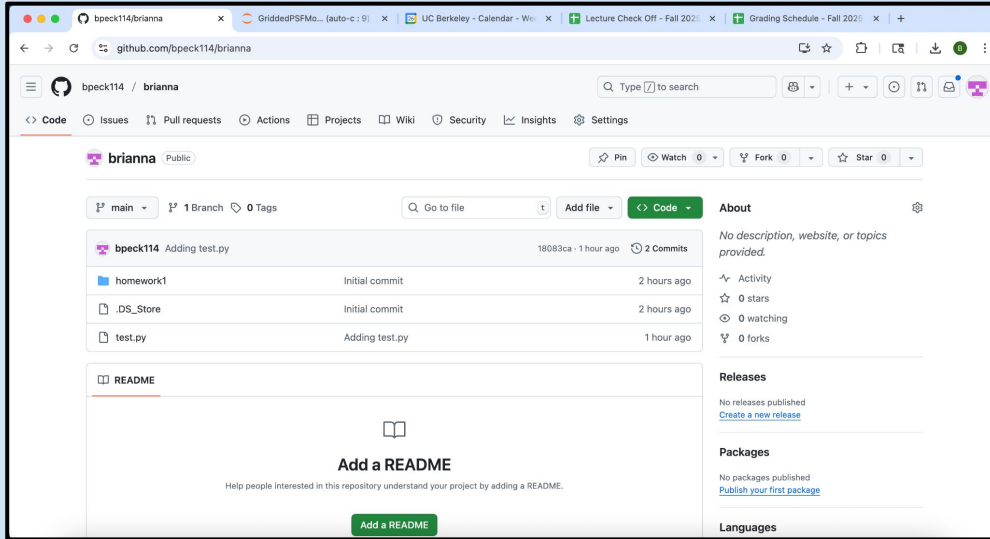
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Adding test.py"
[main 18083ca] Adding test.py
 1 file changed, 2 insertions(+)
 create mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 346 bytes | 346.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:bpeck114/brianna.git
 b03c2ac..18083ca main -> main
bpeck@Briannas-MacBook-Pro brianna %
```

REVIEW: GITHUB (REMOTE)

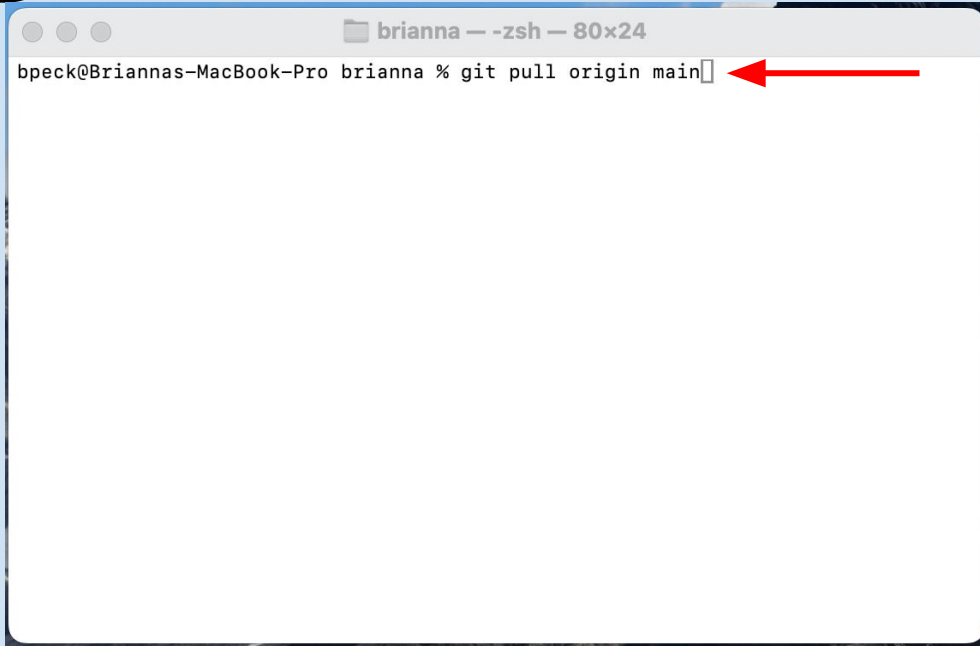
Once you have saved your local changes, you can see them on the internet

WARNING!

It is possible to make changes on the internet, if you then try to save your local changes. You will get an error!



REVIEW: GITHUB (Error)

A terminal window titled 'brianna - zsh - 80x24' showing the command 'bpeck@Briannas-MacBook-Pro brianna % git pull origin main' with a red arrow pointing to the end of the command line.

```
brianna - zsh - 80x24
bpeck@Briannas-MacBook-Pro brianna % git pull origin main
```

To avoid said error. Before making ANY changes locally. Call:

git pull origin main

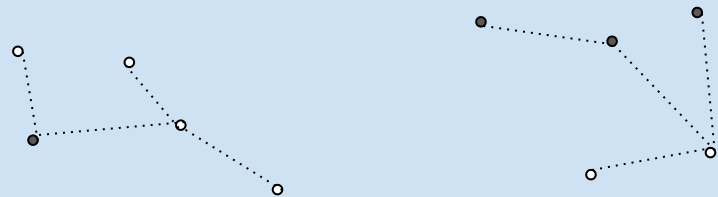
You will learn more about git pull in a future homework!



★ 04 ★

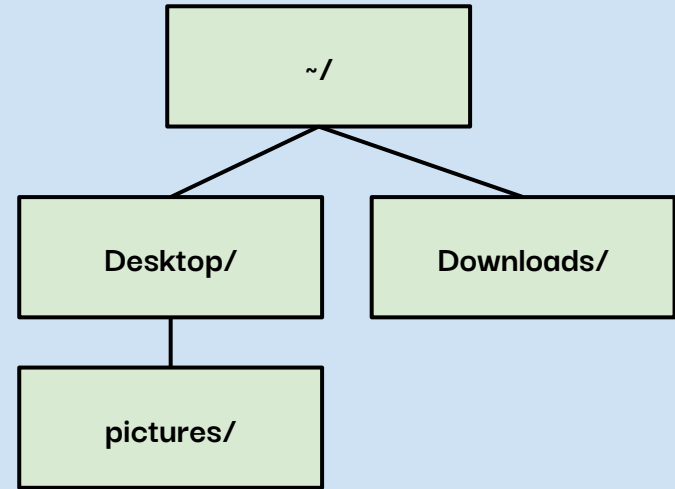
Questions

Directory Trees + File Paths



Learning ABOUT FILE PATHS

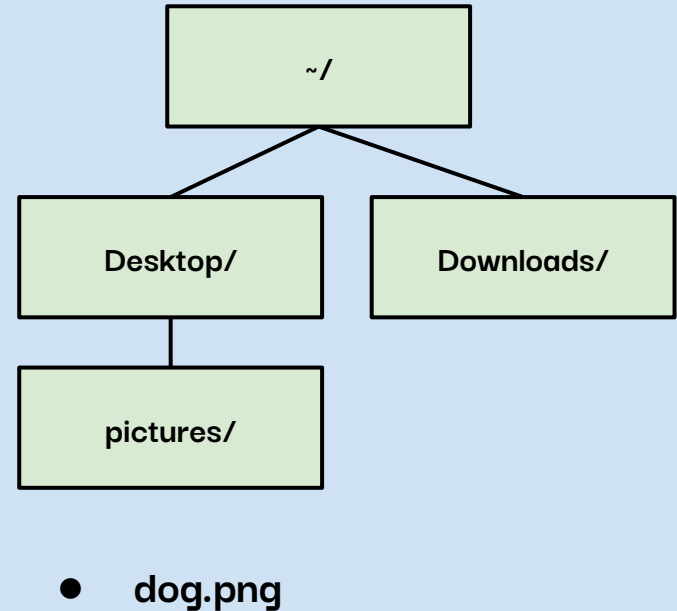
- 1) What is this to the right?
- 2) What is ~/ ?
- 3) What does / after Desktop, Downloads and pictures mean?
- 4) Why is dog.png not in a green box?



- dog.png

Answers to questions

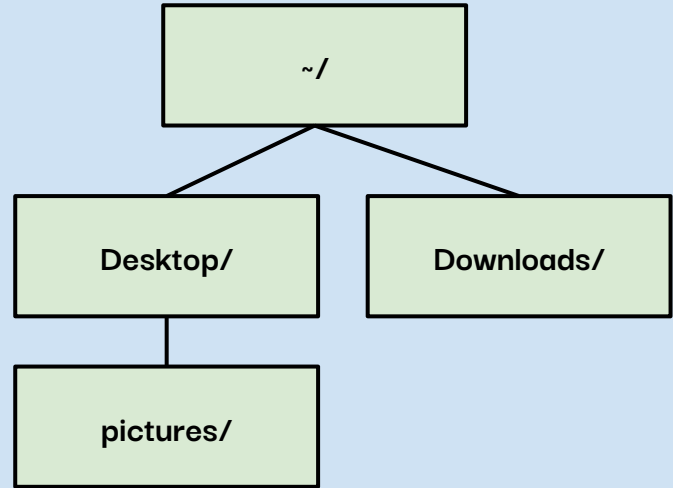
- 1) Directory tree
- 2) Home directory
- 3) It's a directory/folder
- 4) Green box = folder
Dot = file (dog.png)



create a Directory tree

Create a directory tree of your python_decal_sp26 directory:

- Directories = boxes
- Files = dots
- Lines show what is in each folder



- dog.png

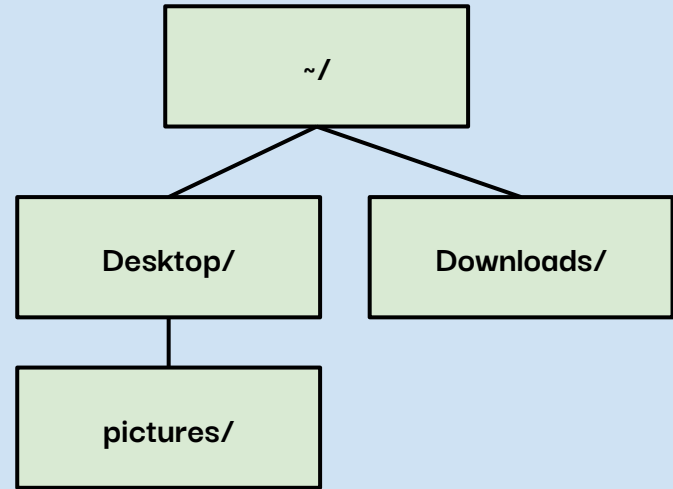
Learning ABOUT FILE PATHS

Start in the folder: pictures/

- Write an absolute path to Downloads/
- Write a relative path to Downloads/

What command would you call to change to that directory?

•



- dog.png

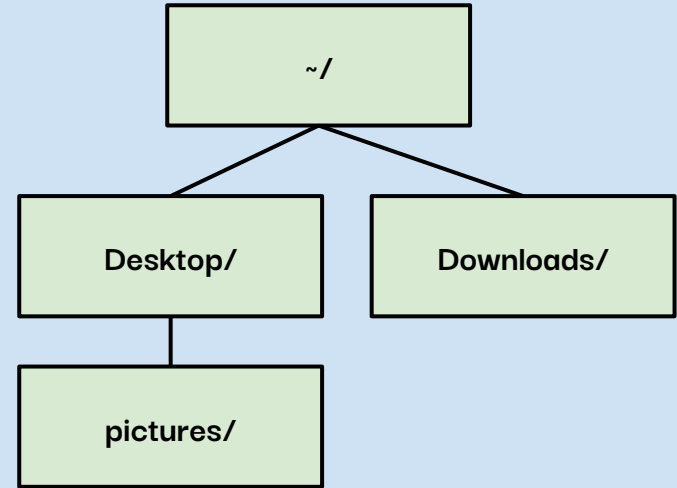
Answers to questions

Start in the folder: pictures/

- ~/Downloads
- ../../Downloads

cd ~/Downloads OR

cd ../../Downloads



- dog.png



MORE QUESTIONS



- 1) What is the difference between a local and remote repository?
- 2) What's the difference between Git and GitHub?
- 3) How do you use Python on the terminal?





Answers TO QUESTIONS



- 1) Local = exists on your computer, you can make changes/add files/commit code, managed using Git (git add, git commit)

Remote = hosted online with GitHub, allows you to collaborate and backup code, you use push and pull commands

- 2) Git = a version control system, tracks changes in your code over time, works locally on your computer, lets you commit/undo changes

GitHub = a website that hosts Git repositories online, lets you store/share/collaborate on projects

- 3) Call:

- python3 <file_name>
 - Runs the code in python, prints it on the terminal
- python3

- - Then you access a python terminal
- 
- 
- 



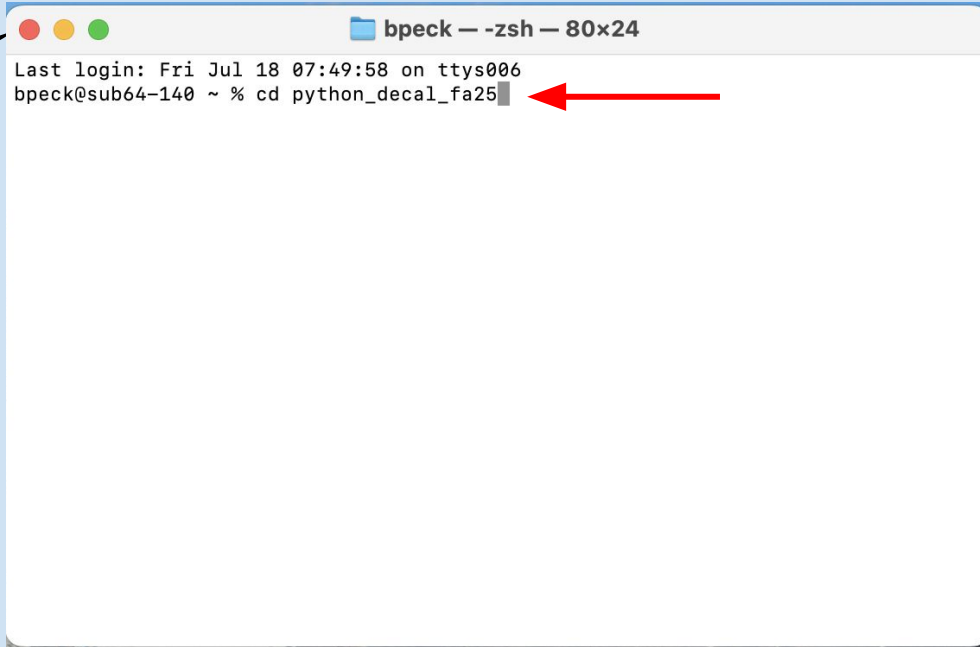
BIG QUESTION



In `python_decal_sp26`, make a folder called lecture_notes/

- Move datatypes.py and script.py to lecture_notes/
 - *Hint: Use the mv command.*
 - Move the file test.py in yourname/ to lecture_notes/
 - Save your changes in yourname/
 - In test.py, write 5 math expressions with these operators:
 - Addition, subtraction, multiplication, division, exponent
 - Run your code on the terminal
 - Run your code on VS Code
 -
- 
- 
- 

OPEN THE Terminal + MOVE

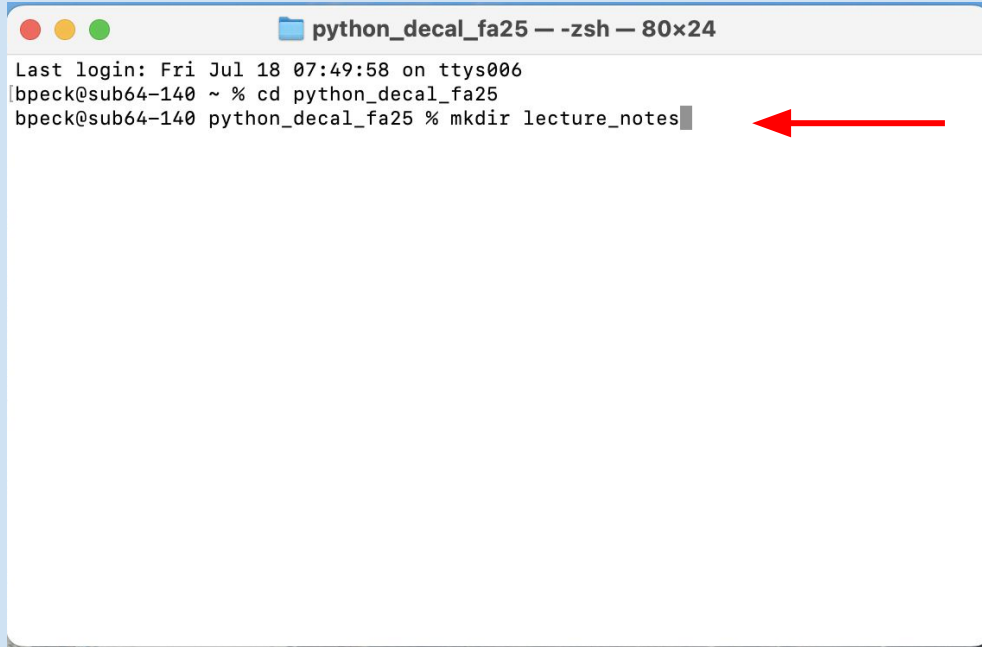
A screenshot of a terminal window titled "bpeck — zsh — 80x24". The window shows the output of a login session: "Last login: Fri Jul 18 07:49:58 on ttys006" followed by the prompt "bpeck@sub64-140 ~ %". The user has entered the command "cd python_decal_fa25", and a red arrow points to the end of the command line.

```
bpeck — zsh — 80x24
Last login: Fri Jul 18 07:49:58 on ttys006
bpeck@sub64-140 ~ % cd python_decal_fa25
```

Open the terminal.

Move into your
python_decal_sp26 folder

Make a New Directory



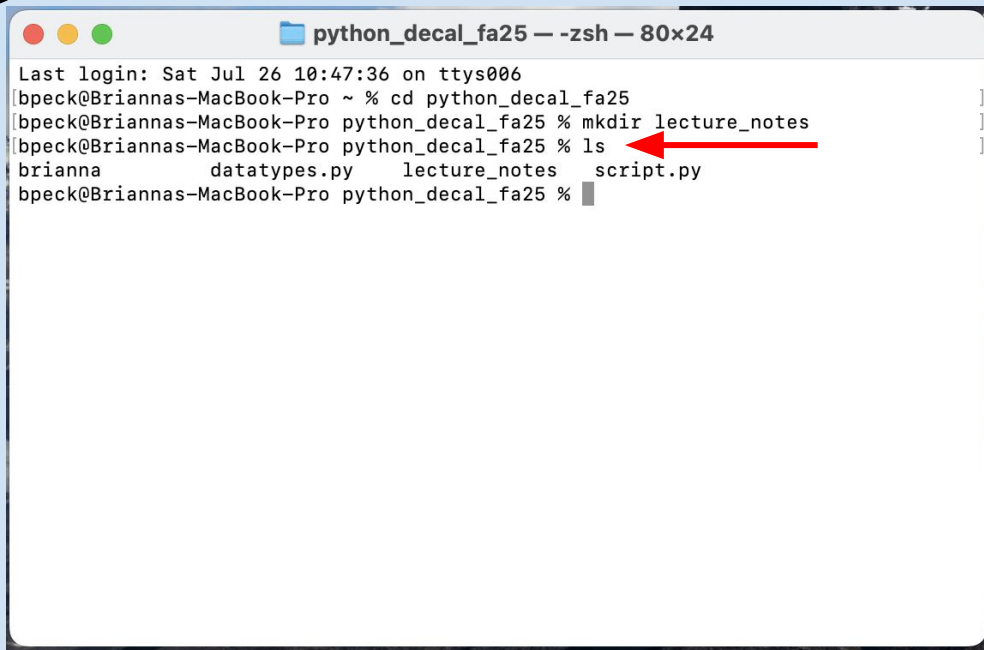
```
python_decal_fa25 — -zsh — 80x24
Last login: Fri Jul 18 07:49:58 on ttys006
bpeck@sub64-140 ~ % cd python_decal_fa25
bpeck@sub64-140 python_decal_fa25 % mkdir lecture_notes
```

A terminal window titled "python_decal_fa25 — -zsh — 80x24" showing a sequence of commands. The first command is "cd python_decal_fa25" and the second is "mkdir lecture_notes". A red arrow points to the "mkdir lecture_notes" command.

To make a new directory,
call: **mkdir lecture_notes**

Today's lecture check code:
yellow brick road

LIST contents

A terminal window titled 'python_decal_fa25 - zsh - 80x24' showing a series of commands and their outputs. The commands are: 'cd python_decal_fa25', 'mkdir lecture_notes', and 'ls'. The output of 'ls' shows three items: 'brianna', 'datatypes.py', and 'lecture_notes script.py'. A red arrow points to the 'lecture_notes' entry in the output.

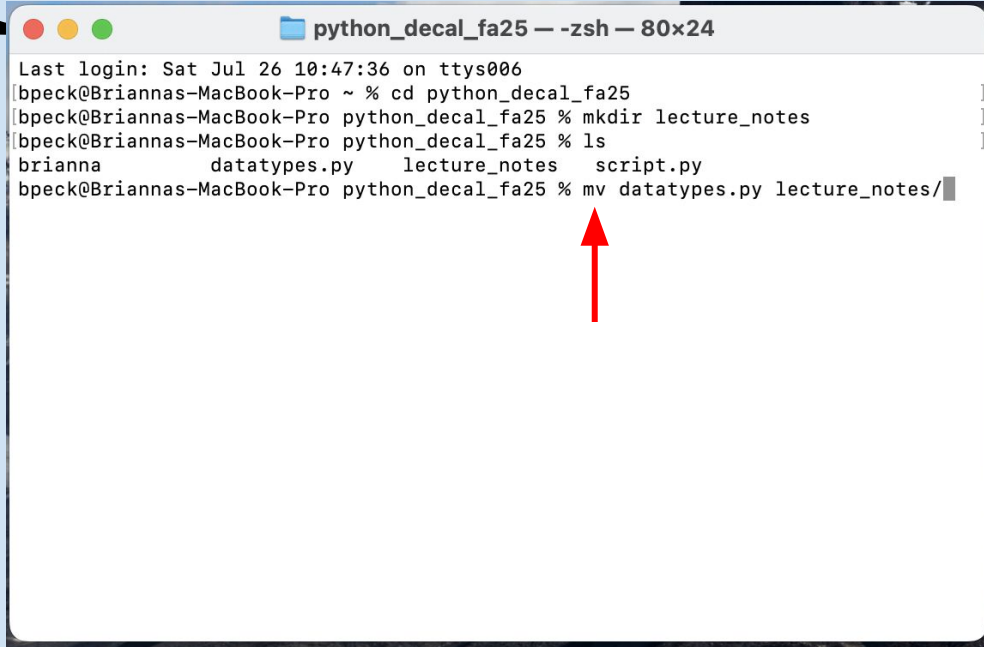
```
python_decal_fa25 — -zsh — 80x24
Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 %
```

To see our new folder, we can call: ls

At minimum, you should have:

- brianna/
- lecture_notes/
- datatypes.py
- script.py

MOVE FILE

A terminal window titled 'python_decal_fa25 — -zsh — 80x24' with a blue folder icon. The window shows a series of commands and their outputs. A red arrow points to the 'mv' command in the last line.

```
Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv datatypes.py lecture_notes/
```

To move a file, we will use the mv command

The syntax is:

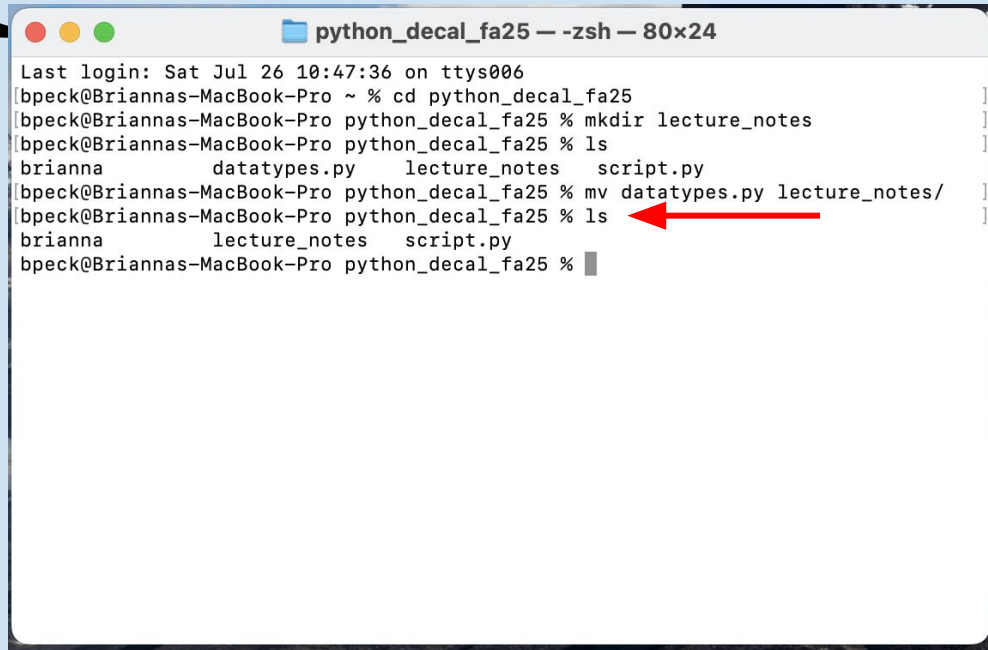
mv <file name> <new folder>

So:

file name = datatypes.py

new folder = lecture_notes/

LIST contents

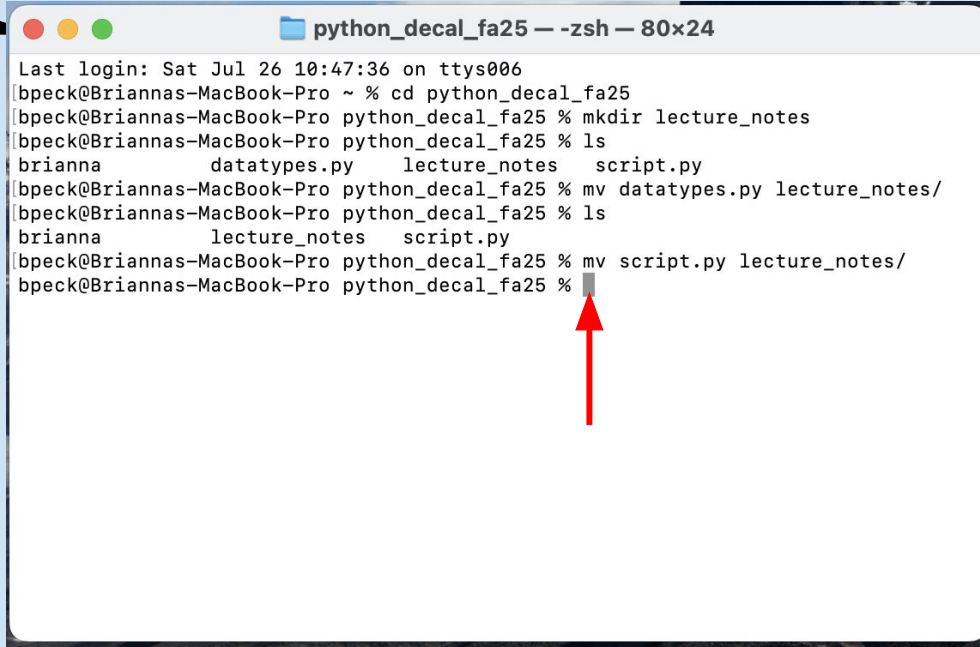
A terminal window titled 'python_decal_fa25 — zsh — 80x24' with a blue title bar and red, yellow, and green window control buttons. The terminal shows a series of commands and their outputs. A red arrow points to the output of the 'ls' command, which lists 'brianna', 'lecture_notes', and 'script.py'.

```
Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 %
```

Calling ls again will show us that datatypes.py has disappeared.

Let's repeat this command again but with script.py

MOVE FILE

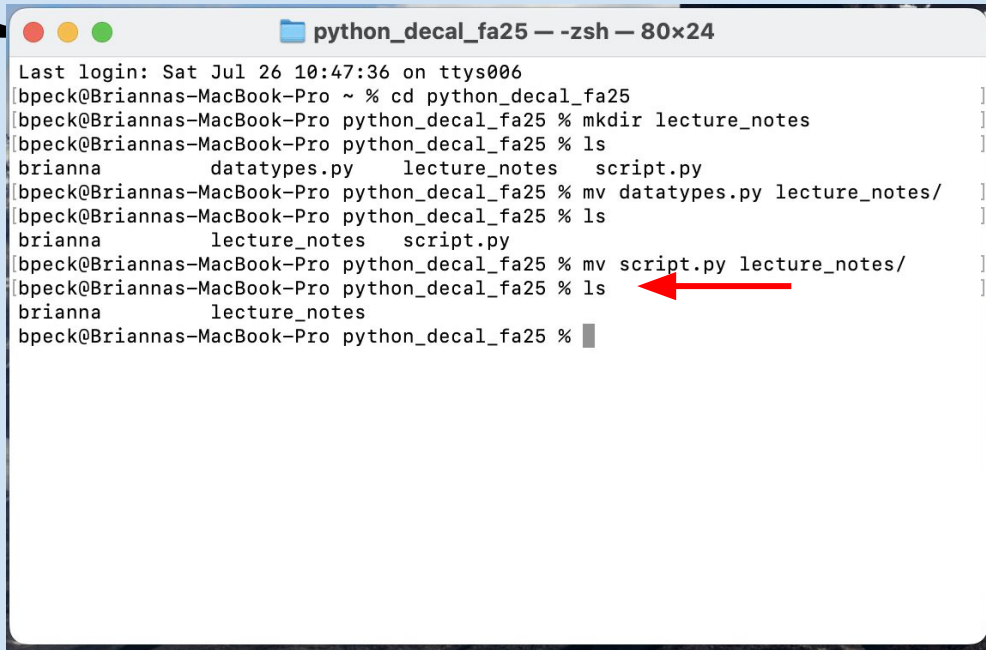
A terminal window titled 'python_decal_fa25 — zsh — 80x24' with standard macOS window controls (red, yellow, green buttons). The terminal shows a series of commands and their outputs. The final command, 'mv script.py lecture_notes/', is being entered, with a red arrow pointing to the cursor. The terminal text is as follows:

```
Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv script.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 %
```

So we will type:

mv script.py lecture_notes/

LIST contents

A terminal window titled 'python_decal_fa25 — zsh — 80x24' with a blue title bar and red, yellow, and green window control buttons. The terminal shows a series of commands and their outputs. A red arrow points to the output of the 'ls' command after moving 'script.py' to the 'lecture_notes' directory.

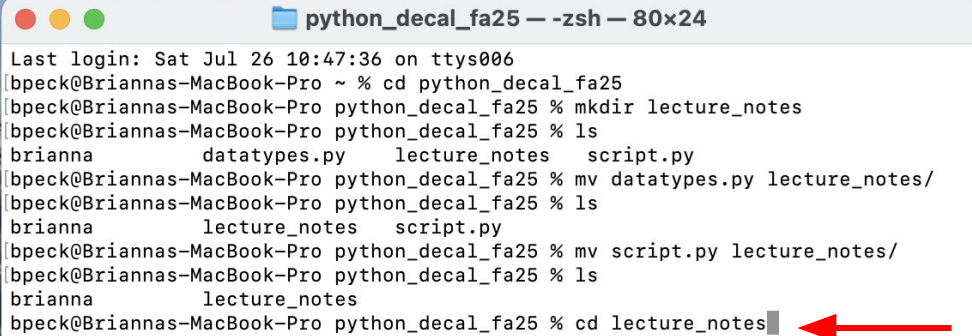
```
Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv script.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 %
```

Calling ls again will show both the files, are now no longer in python_decal_sp26

Let's move into lecture_notes to see the new files

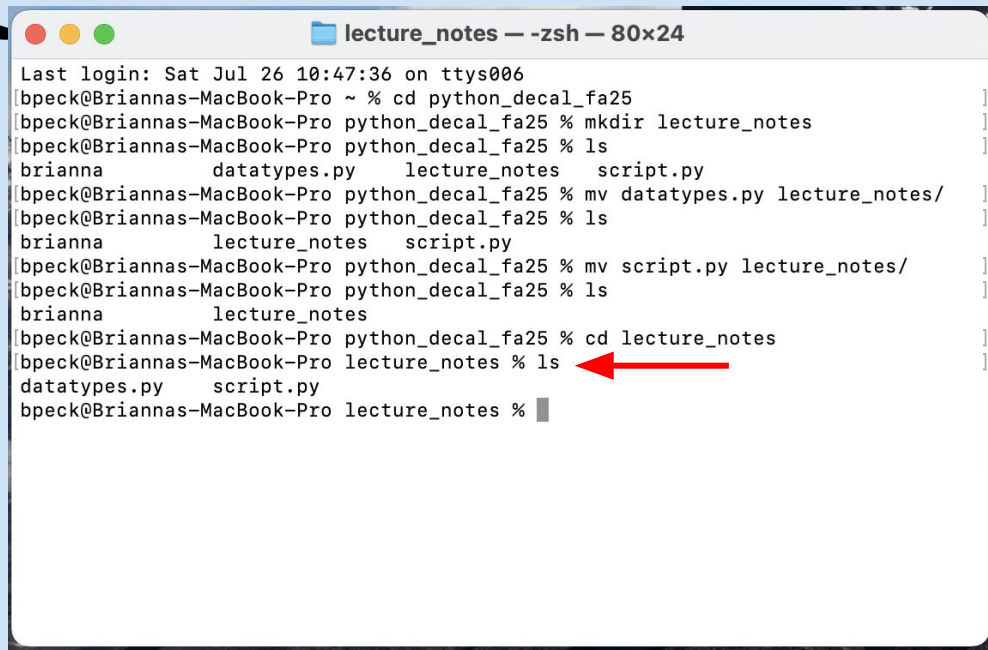
MOVING DIRECTORIES

To move directories, we will
call `cd lecture_notes/`

A terminal window titled 'python_decal_fa25 — zsh — 80x24' with a blue title bar and red, yellow, and green window control buttons. The terminal shows a series of commands and their outputs. A red arrow points to the final command 'cd lecture_notes'.

```
python_decal_fa25 — zsh — 80x24
Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv script.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd lecture_notes
```

LIST CONTENTS



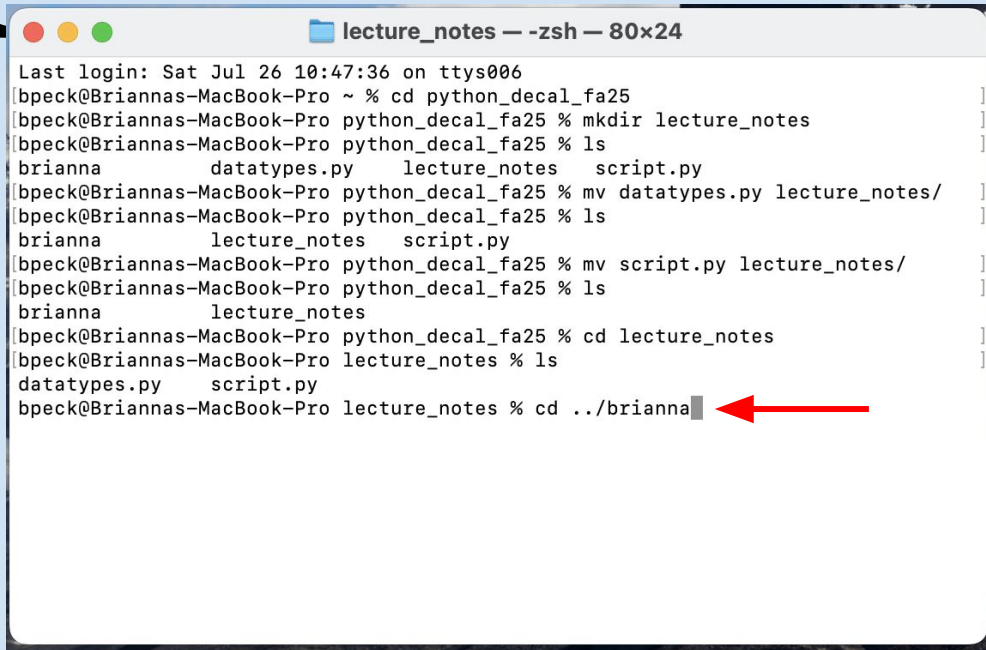
```
lecture_notes — zsh — 80x24

Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decad_fa25
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mv script.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      lecture_notes
bpeck@Briannas-MacBook-Pro python_decad_fa25 % cd lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes % ls
datatypes.py  script.py
bpeck@Briannas-MacBook-Pro lecture_notes %
```

Calling **ls** shows us the two files that we moved over:

- datatypes.py
- script.py

MOVING DIRECTORIES



```
lecture_notes — zsh — 80x24

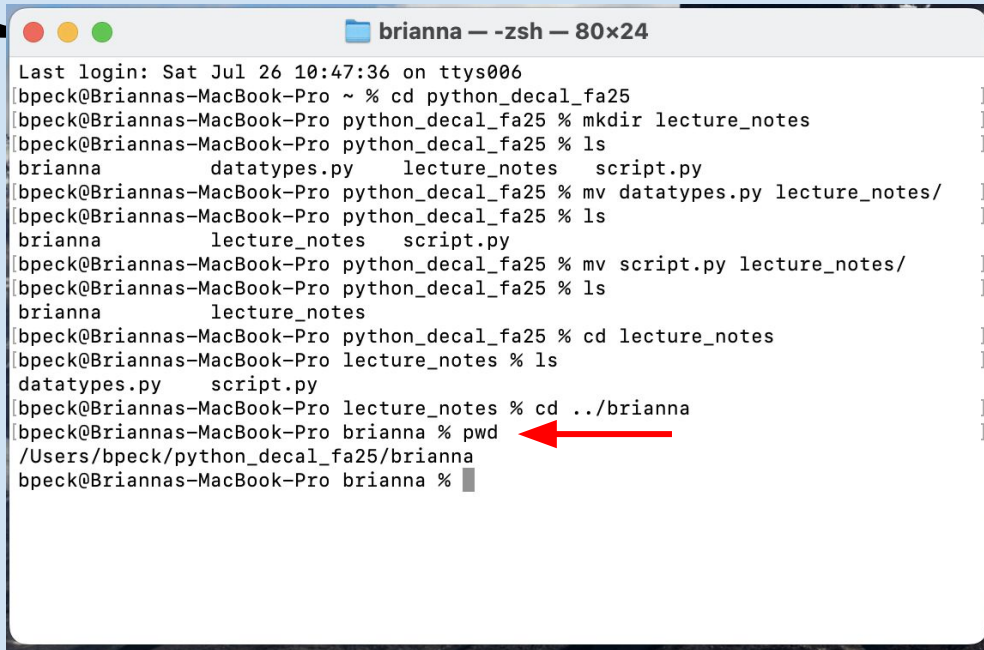
Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv script.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes % ls
datatypes.py  script.py
bpeck@Briannas-MacBook-Pro lecture_notes % cd ../brianna
```



Our next step is to move test.py from last class into our lecture_notes folder.

Let's write a relative path and move immediately into our yourname/ repository

PRINT CWD

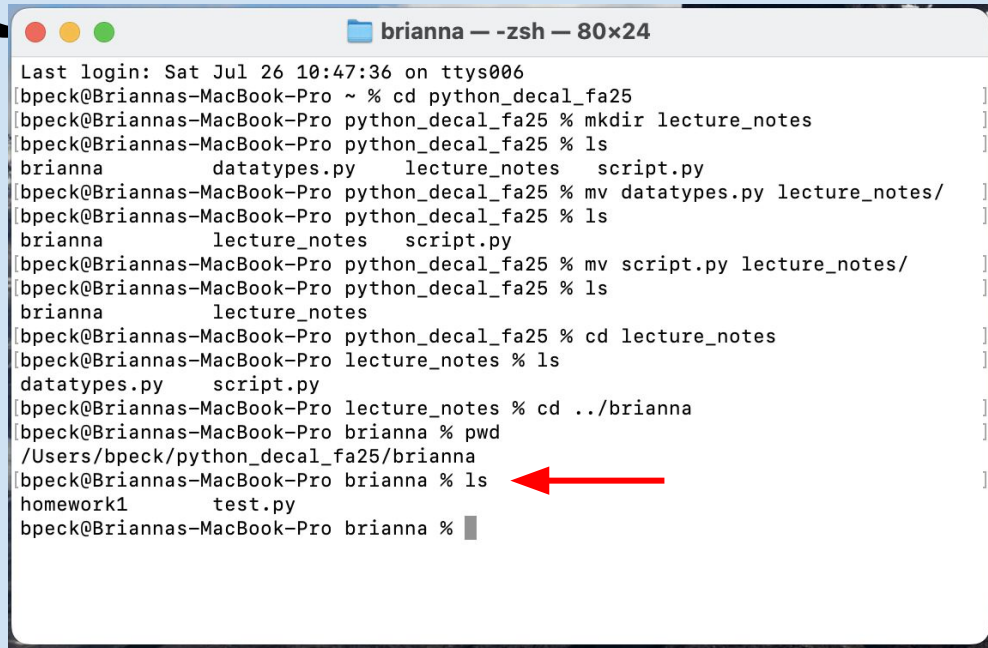


```
brianna — -zsh — 80x24

Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decad_fa25
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mv script.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      lecture_notes
bpeck@Briannas-MacBook-Pro python_decad_fa25 % cd lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes % ls
datatypes.py  script.py
bpeck@Briannas-MacBook-Pro lecture_notes % cd ../brianna
bpeck@Briannas-MacBook-Pro brianna % pwd
/Users/bpeck/python_decad_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna %
```

Calling pwd reveals that we have now moved into the yourname/ folder

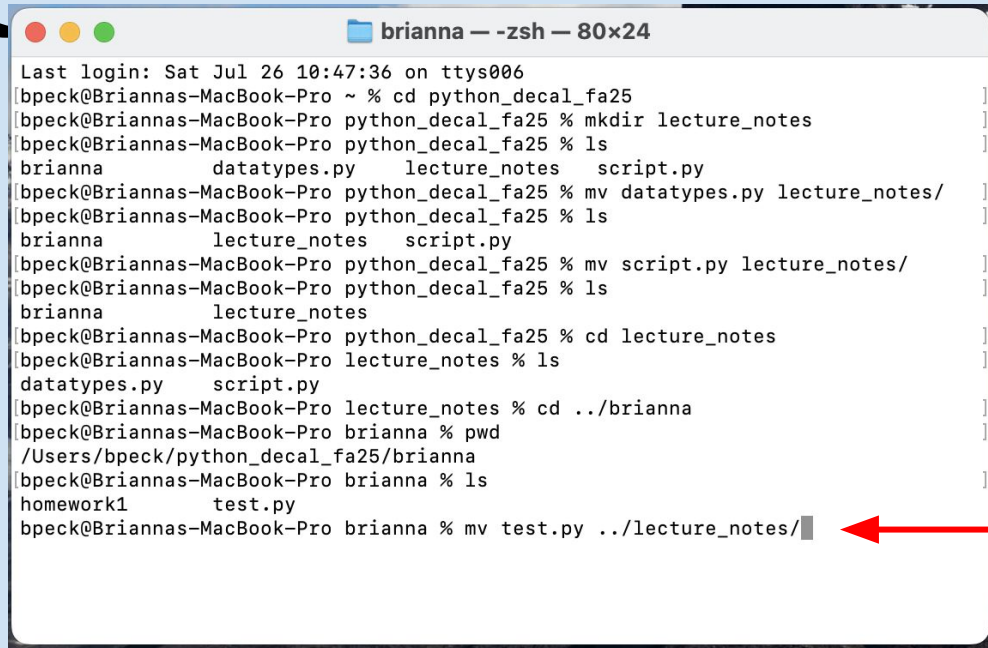
LIST CONTENTS



```
brianna — -zsh — 80x24
Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decad_fa25
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mv script.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      lecture_notes
bpeck@Briannas-MacBook-Pro python_decad_fa25 % cd lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes % ls
datatypes.py  script.py
bpeck@Briannas-MacBook-Pro lecture_notes % cd ../brianna
bpeck@Briannas-MacBook-Pro brianna % pwd
/Users/bpeck/python_decad_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1    test.py
bpeck@Briannas-MacBook-Pro brianna %
```

Calling ls shows us that yourname/ is currently holding the test.py file that we want to move

MOVE FILE

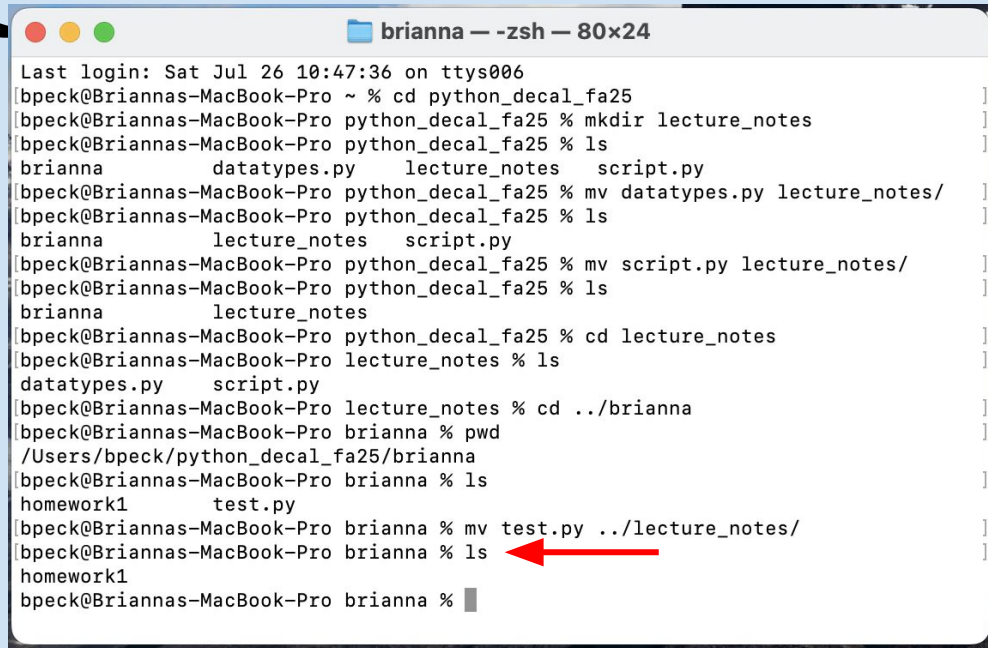
A terminal window titled 'brianna — -zsh — 80x24' with a blue title bar and red, yellow, and green window control buttons. The terminal shows a series of commands and their outputs. A red arrow points to the final command: 'mv test.py ../lecture_notes/'.

```
Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decad_fa25
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mv script.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      lecture_notes
bpeck@Briannas-MacBook-Pro python_decad_fa25 % cd lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes % ls
datatypes.py  script.py
bpeck@Briannas-MacBook-Pro lecture_notes % cd ../brianna
bpeck@Briannas-MacBook-Pro brianna % pwd
/Users/bpeck/python_decad_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1    test.py
bpeck@Briannas-MacBook-Pro brianna % mv test.py ../lecture_notes/
```

To move the file we will use the mv command and write a relative path to lecture_notes/

Super secret code:
yellow brick road

LIST CONTENTS



```
brianna — -zsh — 80x24

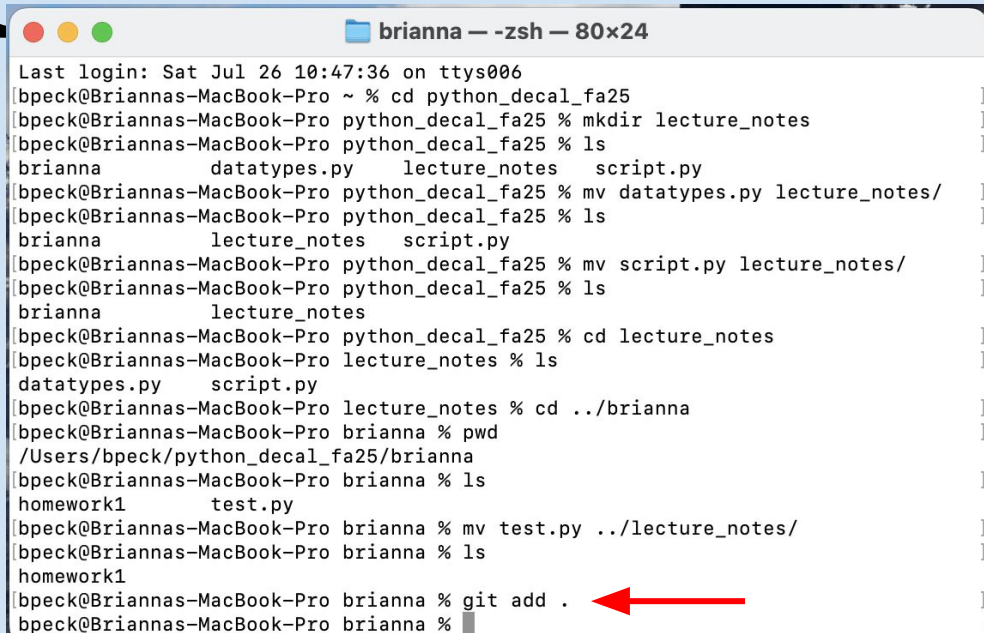
Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv script.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes % ls
datatypes.py  script.py
bpeck@Briannas-MacBook-Pro lecture_notes % cd ../brianna
bpeck@Briannas-MacBook-Pro brianna % pwd
/Users/bpeck/python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1    test.py
bpeck@Briannas-MacBook-Pro brianna % mv test.py ../lecture_notes/
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna %
```

Calling ls shows us that the file is now gone!

Let's save our changes by running through the Git procedure:

- git add
- git commit
- git push

ADD CHANGES



```
brianna — -zsh — 80x24
Last login: Sat Jul 26 10:47:36 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decad_fa25
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mkdir lecture_notes
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decad_fa25 % mv script.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decad_fa25 % ls
brianna      lecture_notes
bpeck@Briannas-MacBook-Pro python_decad_fa25 % cd lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes % ls
datatypes.py  script.py
bpeck@Briannas-MacBook-Pro lecture_notes % cd ../brianna
bpeck@Briannas-MacBook-Pro brianna % pwd
/Users/bpeck/python_decad_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1    test.py
bpeck@Briannas-MacBook-Pro brianna % mv test.py ../lecture_notes/
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna %
```

Recall: yourname/ tracks the changes in the folder

But we need to tell the repository when we have made a change

Start with: git add .

commit changes

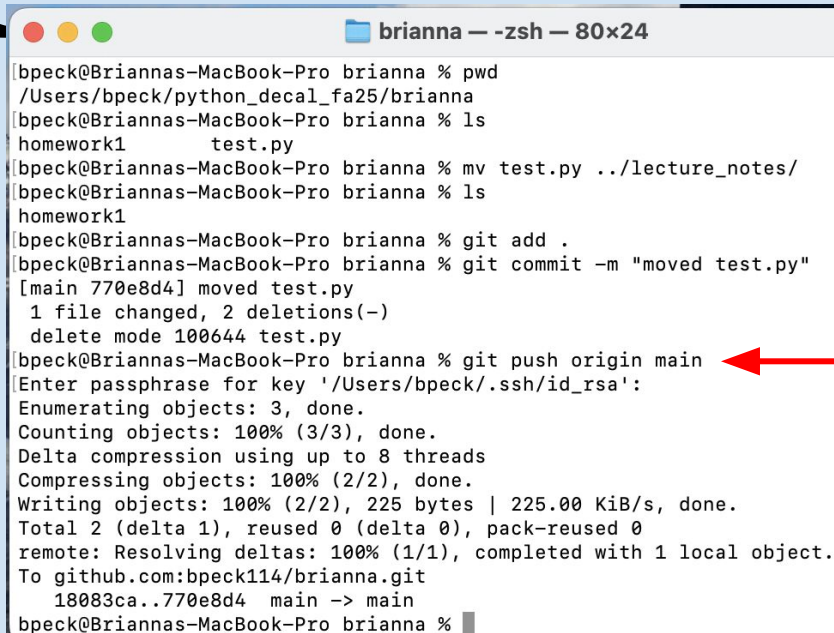
```
brianna      datatypes.py  lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv datatypes.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes  script.py
bpeck@Briannas-MacBook-Pro python_decal_fa25 % mv script.py lecture_notes/
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna      lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes % ls
datatypes.py  script.py
bpeck@Briannas-MacBook-Pro lecture_notes % cd ../brianna
bpeck@Briannas-MacBook-Pro brianna % pwd
/Users/bpeck/python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1    test.py
bpeck@Briannas-MacBook-Pro brianna % mv test.py ../lecture_notes/
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "moved test.py"
[main 770e8d4] moved test.py
 1 file changed, 2 deletions(-)
 delete mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna %
```

Next command is:

git commit -m "moved test.py"

This is committing our “move”
which looks like a “deletion”
to Git

PUSH CHANGES

A terminal window titled 'brianna — -zsh — 80x24' showing a series of commands and their outputs. The commands include 'pwd', 'ls', 'mv', 'git add', 'git commit -m "moved test.py"', and 'git push origin main'. A red arrow points to the 'git push origin main' command. The output shows the commit process, including file changes, deletions, and the final push to the remote repository.

```
bpeck@Briannas-MacBook-Pro brianna % pwd
/Users/bpeck/python_decad_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1      test.py
bpeck@Briannas-MacBook-Pro brianna % mv test.py ../lecture_notes/
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "moved test.py"
[main 770e8d4] moved test.py
 1 file changed, 2 deletions(-)
 delete mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 225 bytes | 225.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:bpeck114/brianna.git
 18083ca..770e8d4  main -> main
bpeck@Briannas-MacBook-Pro brianna %
```

Finally, to save our changes
remotely we will call:
git push origin main

MOVE DIRECTORIES

```
lecture_notes — -zsh — 80x24
/Users/bpeck/python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1      test.py
bpeck@Briannas-MacBook-Pro brianna % mv test.py ../lecture_notes/
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "moved test.py"
[main 770e8d4] moved test.py
 1 file changed, 2 deletions(-)
 delete mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 225 bytes | 225.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:bpeck114/brianna.git
 18083ca..770e8d4  main -> main
bpeck@Briannas-MacBook-Pro brianna % cd ../lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes %
```

Let's go back to
lecture_notes/ by writing a
relative path with cd

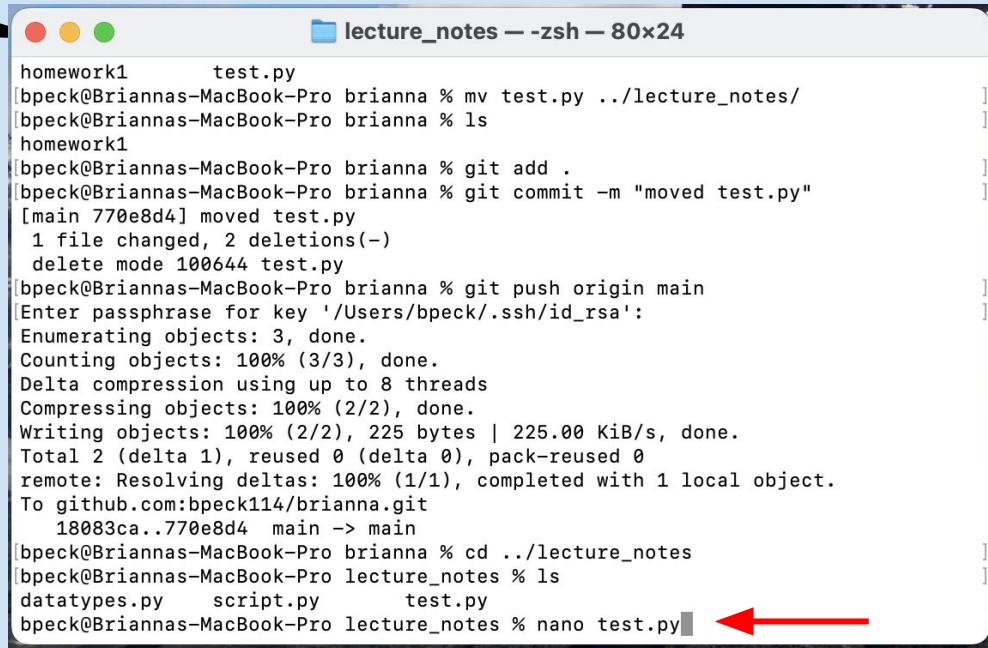
LIST CONTENTS

```
lecture_notes — -zsh — 80x24

homework1      test.py
bpeck@Briannas-MacBook-Pro brianna % mv test.py ../lecture_notes/
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "moved test.py"
[main 770e8d4] moved test.py
 1 file changed, 2 deletions(-)
 delete mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 225 bytes | 225.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:bpeck114/brianna.git
 18083ca..770e8d4  main -> main
bpeck@Briannas-MacBook-Pro brianna % cd ../lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes % ls
datatypes.py  script.py      test.py
bpeck@Briannas-MacBook-Pro lecture_notes %
```

Calling ls shows us that the test.py file is now in lecture_notes/

USE NANO TO EDIT A FILE

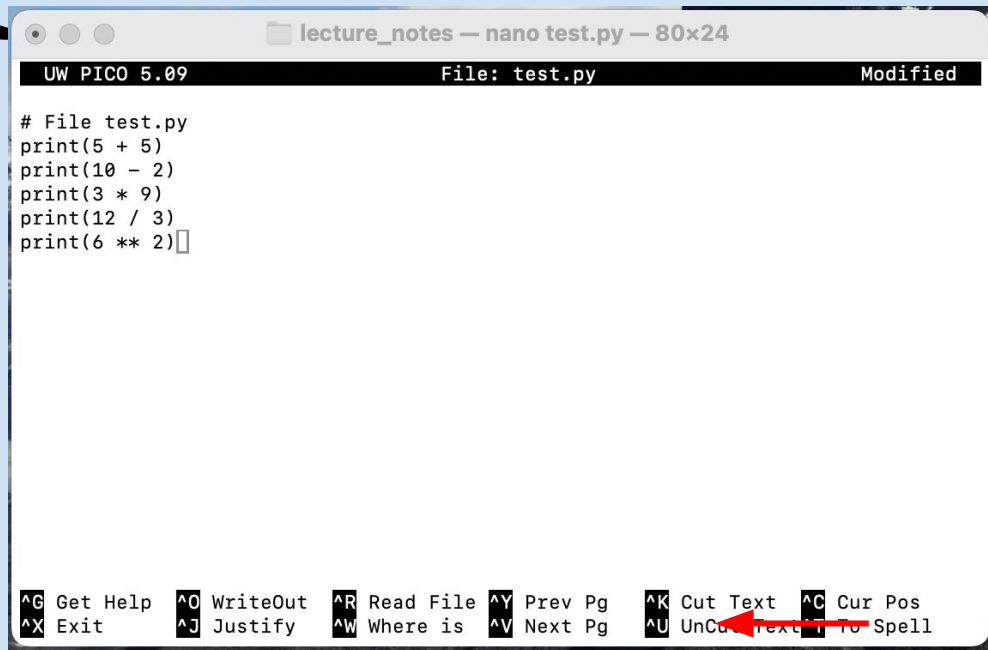


```
lecture_notes — zsh — 80x24

homework1      test.py
bpeck@Briannas-MacBook-Pro brianna % mv test.py ../lecture_notes/
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "moved test.py"
[main 770e8d4] moved test.py
 1 file changed, 2 deletions(-)
 delete mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 225 bytes | 225.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:bpeck114/brianna.git
 18083ca..770e8d4  main -> main
bpeck@Briannas-MacBook-Pro brianna % cd ../lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes % ls
datatypes.py  script.py      test.py
bpeck@Briannas-MacBook-Pro lecture_notes % nano test.py
```

To edit our file, we will call
nano test.py

USE NANO TO EDIT A FILE



```
lecture_notes — nano test.py — 80x24
UW PICO 5.09 File: test.py Modified

# File test.py
print(5 + 5)
print(10 - 2)
print(3 * 9)
print(12 / 3)
print(6 ** 2)

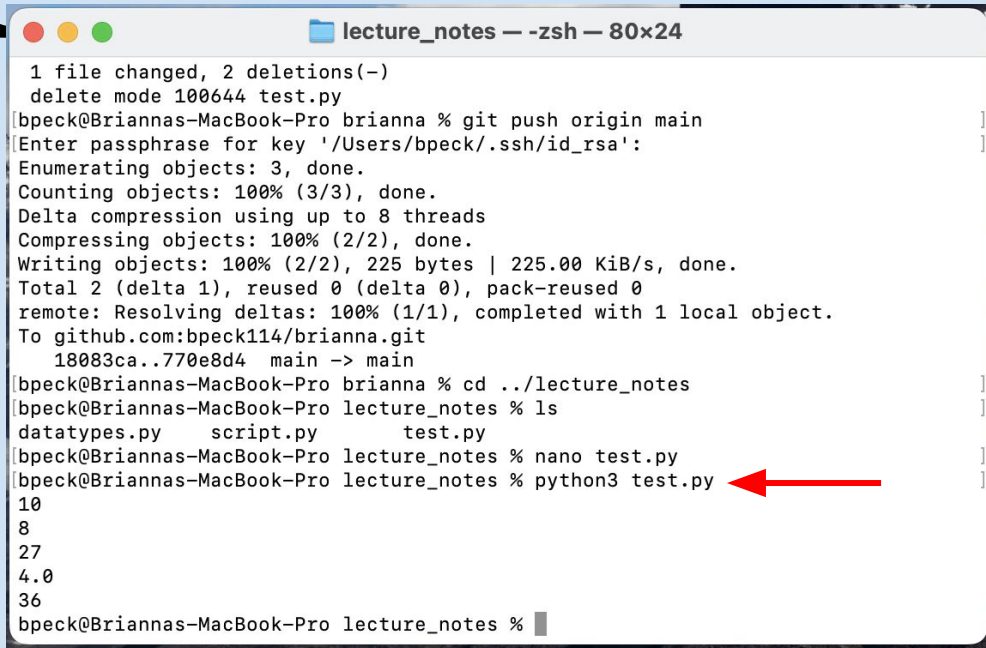
^G Get Help ^O WriteOut ^R Read File ^Y Prev Pg ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where is ^V Next Pg ^U Undo Text ^_ To Spell
```

We are asked to have five operations:

- addition
- subtraction
- multiplication
- division
- exponent

Then, close nano and save your changes to test.py

RUN SCRIPT WITH PYTHON

A terminal window titled 'lecture_notes — zsh — 80x24' showing a series of commands and their outputs. The commands include 'git push origin main', 'cd ../lecture_notes', 'ls', 'nano test.py', and 'python3 test.py'. A red arrow points to the 'python3 test.py' command. The output of the script shows a list of numbers: 10, 8, 27, 4.0, 36.

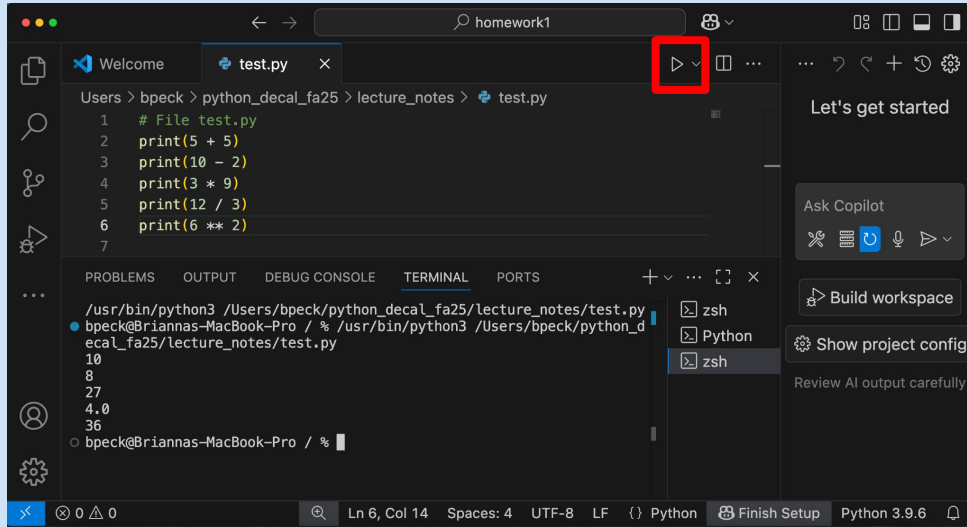
```
1 file changed, 2 deletions(-)
delete mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 225 bytes | 225.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:bpeck114/brianna.git
18083ca..770e8d4  main -> main
bpeck@Briannas-MacBook-Pro brianna % cd ../lecture_notes
bpeck@Briannas-MacBook-Pro lecture_notes % ls
datatypes.py  script.py      test.py
bpeck@Briannas-MacBook-Pro lecture_notes % nano test.py
bpeck@Briannas-MacBook-Pro lecture_notes % python3 test.py
10
8
27
4.0
36
bpeck@Briannas-MacBook-Pro lecture_notes %
```

The main way we have learned
to run scripts on the terminal
is to call:

python3 test.py

RUN SCRIPT WITH VS CODE

To run it on VS Code



```
Users > bpeck > python_decal_fa25 > lecture_notes > test.py
1 # File test.py
2 print(5 + 5)
3 print(10 - 2)
4 print(3 * 9)
5 print(12 / 3)
6 print(6 ** 2)
7
/usr/bin/python3 /Users/bpeck/python_decal_fa25/lecture_notes/test.py
10
8
27
4.0
36
bpeck@Briannas-MacBook-Pro / %
```

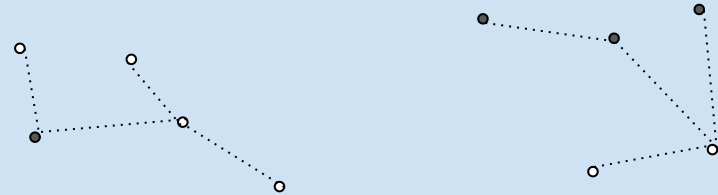
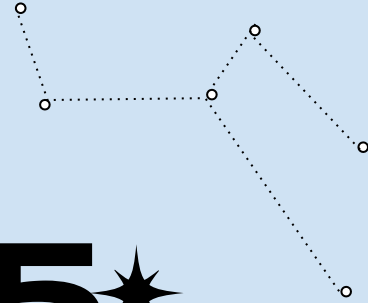
- Open VS Code
- Open the folder `lecture_notes/`
- Open the test.py file
- Press the arrow in the top right-hand corner



★ 05 ★

Lecture Check

Show us what you know!



LECTURE CHECK

Fill out the lecture check [here!](#)

Always due two days after class.

Takes <5 minutes to complete.





★ 06 ★

Bonus Content

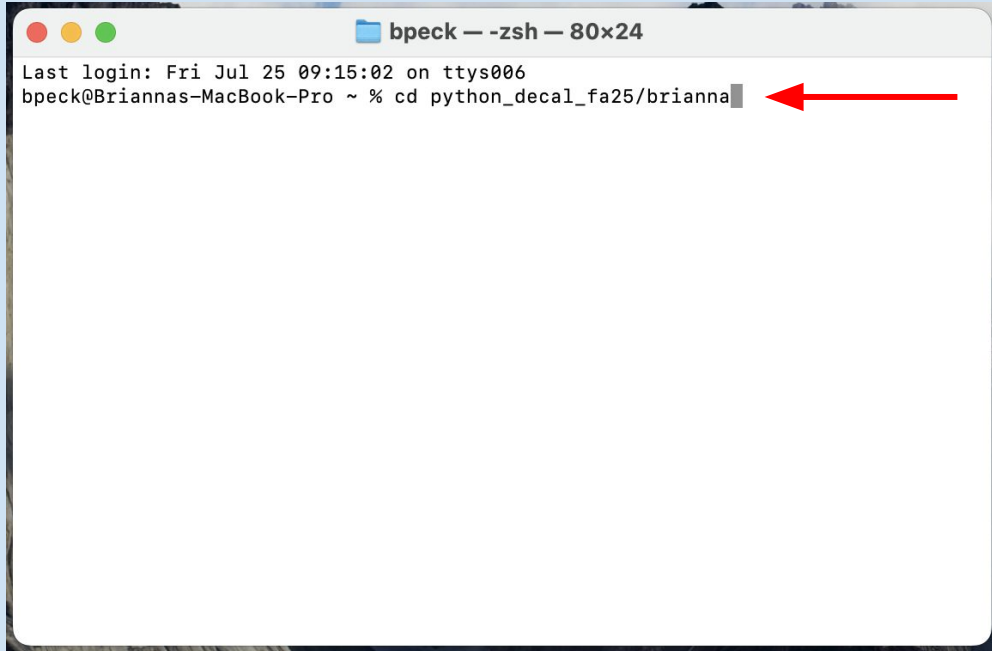
Homework Help





PROBLEM #1

MAKE A NEW FOLDER

A screenshot of a macOS terminal window titled "bpeck — zsh — 80x24". The window shows the output of a login session: "Last login: Fri Jul 25 09:15:02 on ttys006" followed by the prompt "bpeck@Briannas-MacBook-Pro ~ %". The command "cd python_decal_fa25/brianna" has been entered, and a red arrow points to the end of the command line.

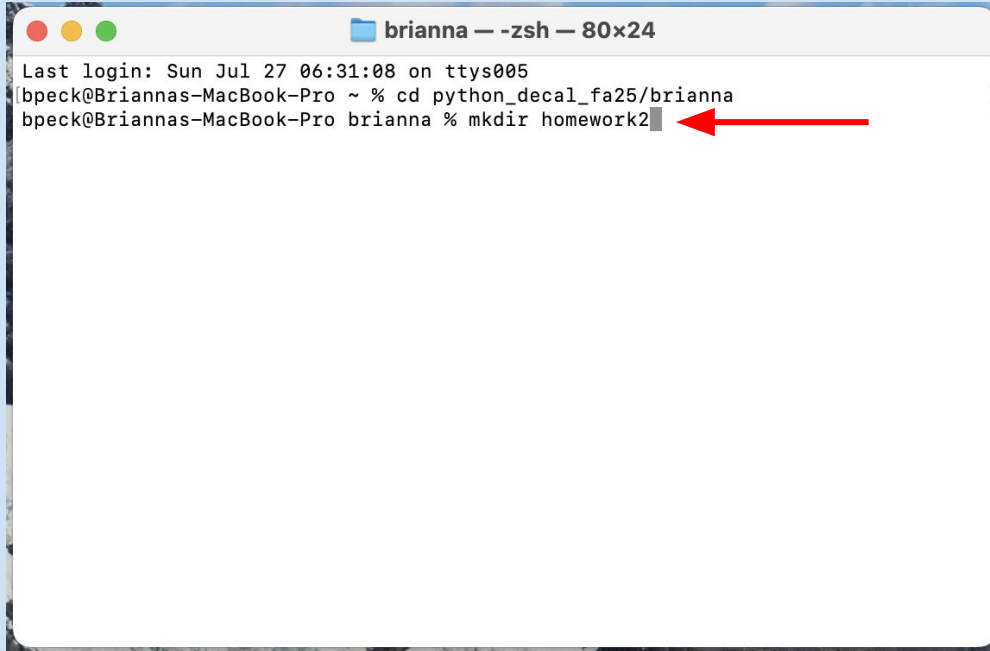
```
bpeck — zsh — 80x24
Last login: Fri Jul 25 09:15:02 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
```

In Problem #1 of HW #2 you are asked to make a new folder called homework2 in the yourname directory.

Start by opening your terminal and navigating to the yourname folder.

MAKE A NEW FOLDER

To make a new folder inside of the yourname folder, type: **mkdir homework2**

A terminal window titled "brianna — -zsh — 80x24" with standard macOS window controls (red, yellow, green buttons). The terminal shows the following text: "Last login: Sun Jul 27 06:31:08 on ttys005", "bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna", and "bpeck@Briannas-MacBook-Pro brianna % mkdir homework2". A red arrow points to the end of the command line.

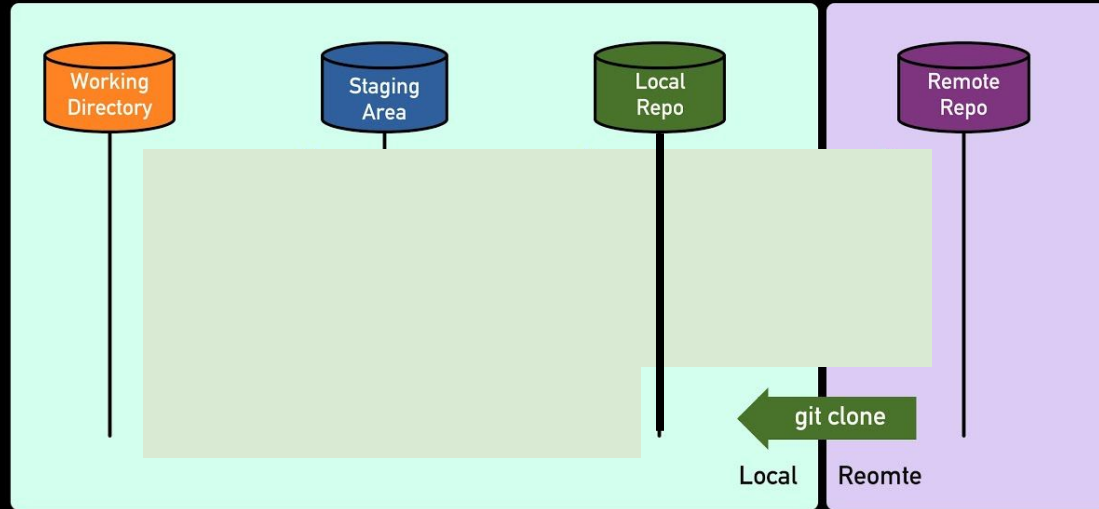
```
brianna — -zsh — 80x24
Last login: Sun Jul 27 06:31:08 on ttys005
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % mkdir homework2
```



PROBLEM #2

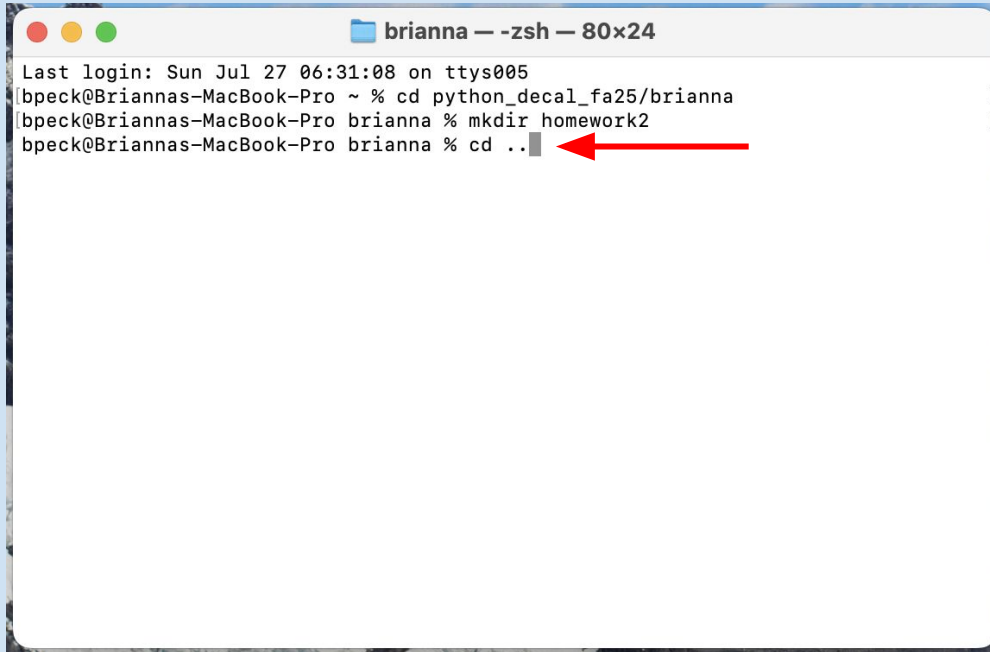
A HELPFUL Diagram

How Git Actually Works



A preview to what git clone does

MOVE TO Parent Directory



```
brianna — -zsh — 80x24
Last login: Sun Jul 27 06:31:08 on ttys005
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % mkdir homework2
bpeck@Briannas-MacBook-Pro brianna % cd ..
```

A terminal window titled "brianna — -zsh — 80x24" showing a series of commands. The first command is "cd python_decal_fa25/brianna", followed by "mkdir homework2", and finally "cd ..". A red arrow points to the "cd .." command, indicating the action of moving to the parent directory.

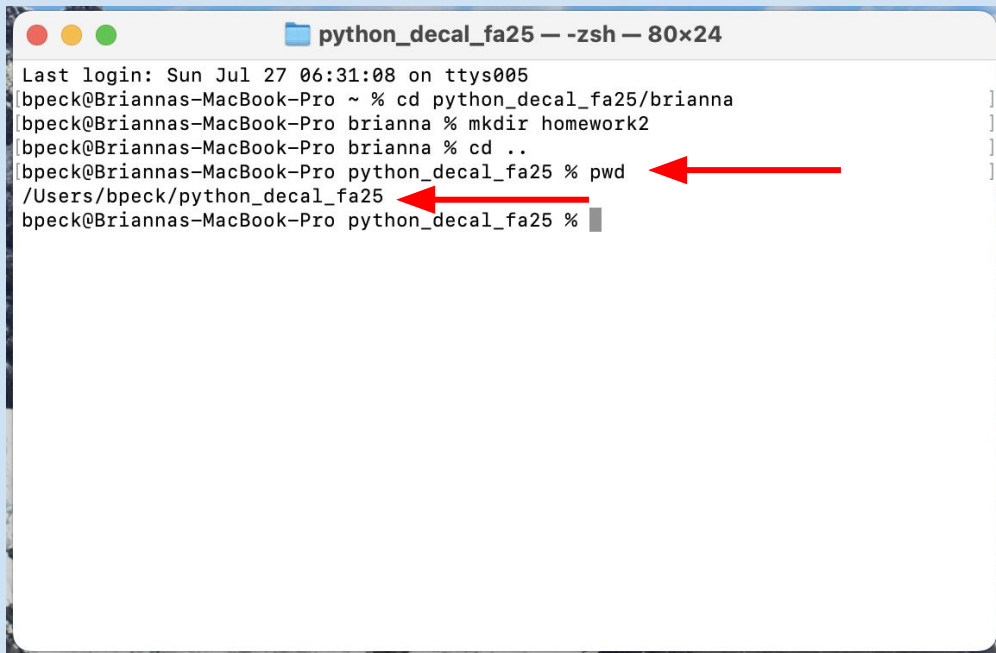
In Problem #2 of HW #2
you are asked to clone a
repository called
course_assignments

From homework2/ move
backup to your parent
directory.

CHECK CWD

Make sure you are in the right location by calling pwd

You should be in python_decal_fa25/



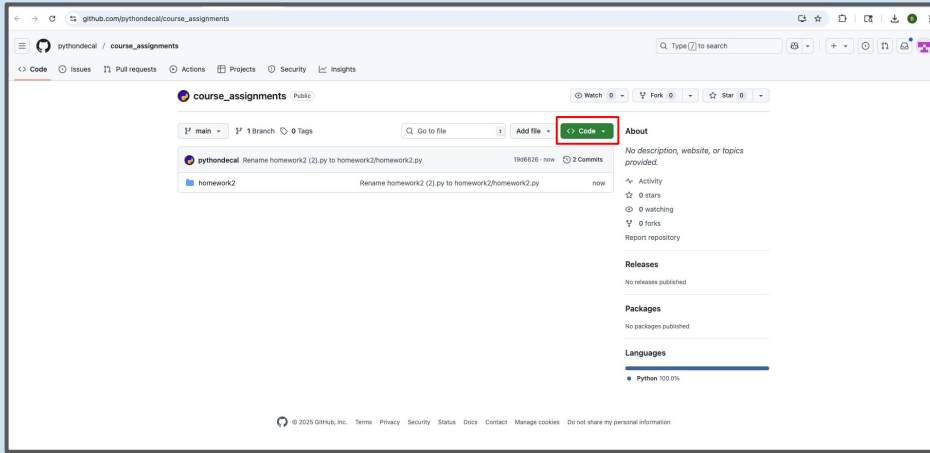
```
python_decal_fa25 — zsh — 80x24
Last login: Sun Jul 27 06:31:08 on ttys005
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % mkdir homework2
bpeck@Briannas-MacBook-Pro brianna % cd ..
bpeck@Briannas-MacBook-Pro python_decal_fa25 % pwd
/Users/bpeck/python_decal_fa25
```

The terminal window shows a sequence of commands: navigating to `python_decal_fa25/brianna`, creating a `homework2` directory, and then navigating back to `python_decal_fa25` using `cd ..`. The `pwd` command is then executed, returning `/Users/bpeck/python_decal_fa25`. Two red arrows point to the `pwd` command and its output to highlight the current working directory.

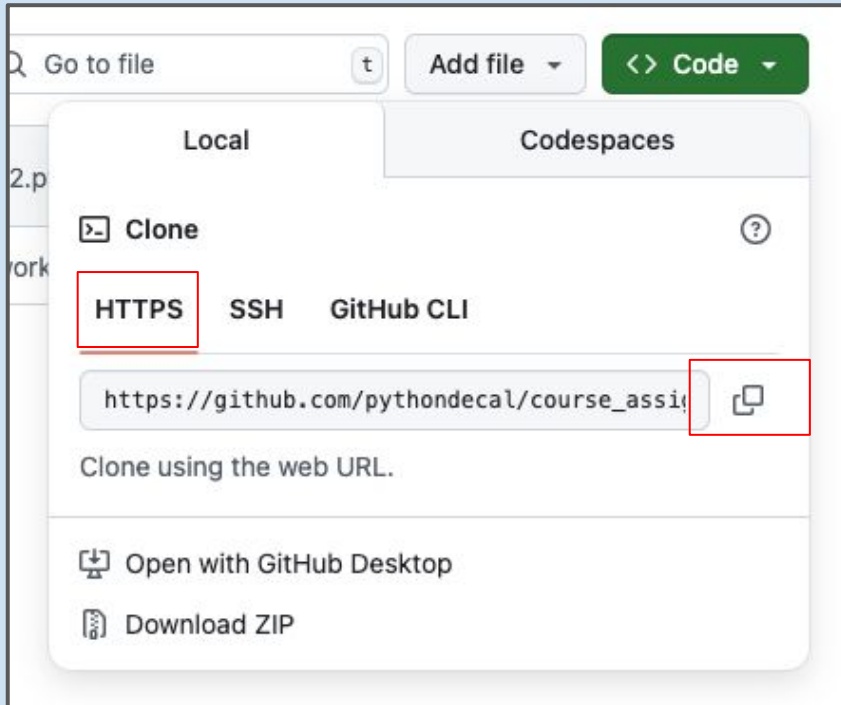
GO TO PYTHONDECAL GITHUB

Click on the link and it will take you to the following site.

Click on the green Code button and make sure you click on the HTTPS link, not the SSH



GO TO PYTHONDECAL GITHUB

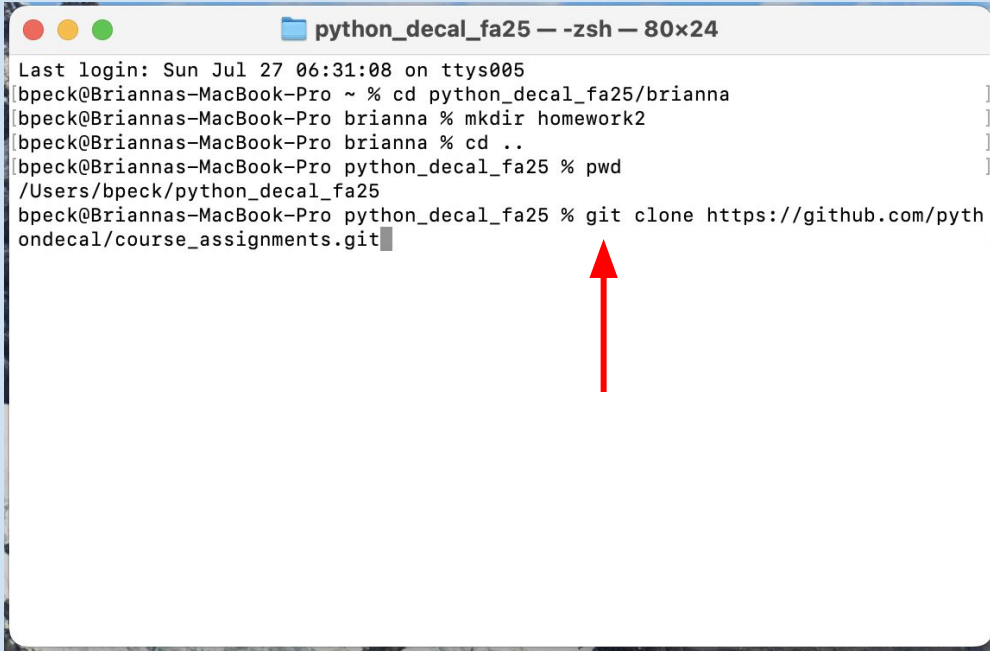


Once you have clicked HTTPS, click on the clipboard icon next to the link

Paste the Link

Go back to your terminal and
type the following:

git clone <paste the link>

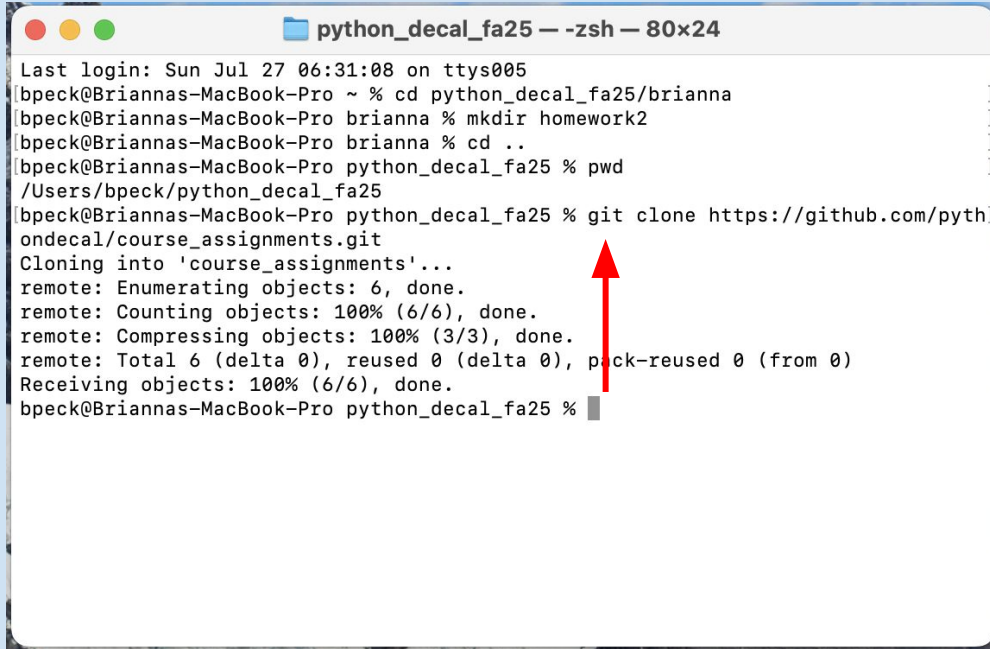


```
python_decal_fa25 — zsh — 80x24
Last login: Sun Jul 27 06:31:08 on ttys005
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % mkdir homework2
bpeck@Briannas-MacBook-Pro brianna % cd ..
bpeck@Briannas-MacBook-Pro python_decal_fa25 % pwd
/Users/bpeck/python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % git clone https://github.com/pyth
ondecad/course_assignments.git
```

Paste the Link

You will get the following output after you paste the link

For this cloned repository you don't not have the ability to change the remote version

A terminal window titled 'python_decad_fa25 — zsh — 80x24' showing a series of commands and their outputs. A red arrow points to the 'git clone' command. The output shows the cloning process, including enumerating, counting, and compressing objects, and finally receiving the objects.

```
python_decad_fa25 — zsh — 80x24
Last login: Sun Jul 27 06:31:08 on ttys005
bpeck@Briannas-MacBook-Pro ~ % cd python_decad_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % mkdir homework2
bpeck@Briannas-MacBook-Pro brianna % cd ..
bpeck@Briannas-MacBook-Pro python_decad_fa25 % pwd
/Users/bpeck/python_decad_fa25
bpeck@Briannas-MacBook-Pro python_decad_fa25 % git clone https://github.com/pyth
ondecad/course_assignments.git
Cloning into 'course_assignments'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
bpeck@Briannas-MacBook-Pro python_decad_fa25 %
```

CONFIRM THE FILE STRUCTURE

The problem also outlines
your file structure should
look like:

```
python_decal_fa25/  
|--- yourname/  
|--- course_assignments/  
|--- lecture_notes/
```

Call ls to confirm this

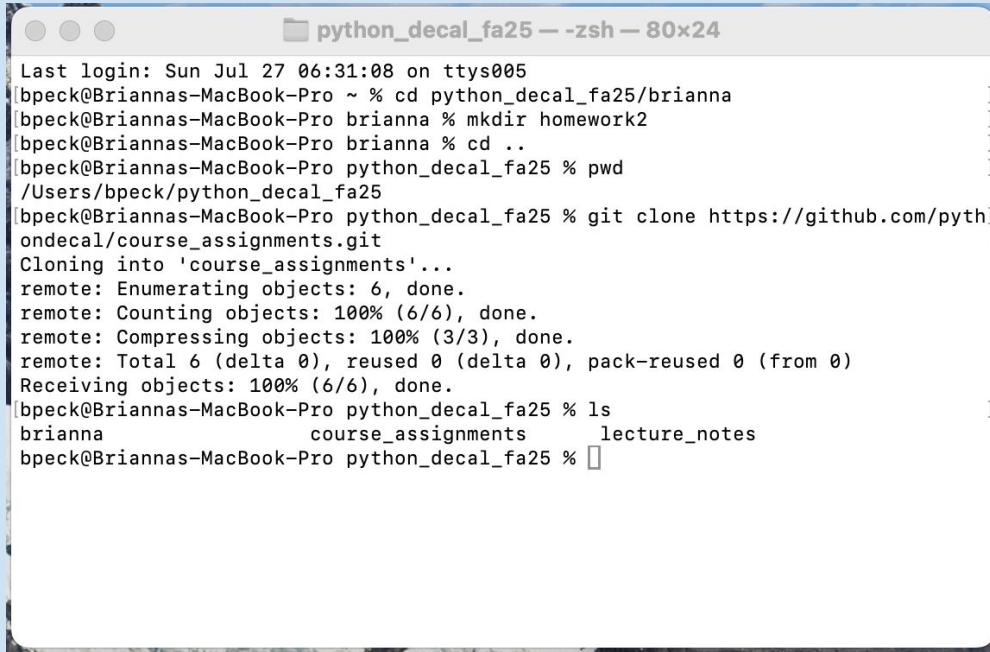
```
python_decal_fa25 — zsh — 80x24  
Last login: Sun Jul 27 06:31:08 on ttys005  
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna  
bpeck@Briannas-MacBook-Pro brianna % mkdir homework2  
bpeck@Briannas-MacBook-Pro brianna % cd ..  
bpeck@Briannas-MacBook-Pro python_decal_fa25 % pwd  
/Users/bpeck/python_decal_fa25  
bpeck@Briannas-MacBook-Pro python_decal_fa25 % git clone https://github.com/pyth  
ondecad/course_assignments.git  
Cloning into 'course_assignments'...  
remote: Enumerating objects: 6, done.  
remote: Counting objects: 100% (6/6), done.  
remote: Compressing objects: 100% (3/3), done.  
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (6/6), done.  
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls  
brianna          course_assignments  lecture_notes  
bpeck@Briannas-MacBook-Pro python_decal_fa25 %
```

Take a screenshot

Take a screenshot of a terminal output like to the left.

Name it: hw2_clone

Save it in your yourname/homework2 folder



```
python_decal_fa25 — zsh — 80x24
Last login: Sun Jul 27 06:31:08 on ttys005
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % mkdir homework2
bpeck@Briannas-MacBook-Pro brianna % cd ..
bpeck@Briannas-MacBook-Pro python_decal_fa25 % pwd
/Users/bpeck/python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % git clone https://github.com/pyth
ondecad/course_assignments.git
Cloning into 'course_assignments'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna          course_assignments  lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 %
```



PROBLEM #2

call cd

In Problem #3 of HW #2
you are asked to copy a file
from a folder in
course_assignments

From python_decal_fa25,
we will move into the
course_assignments
repository

```
course_assignments — zsh — 80x24
Last login: Sun Jul 27 06:31:08 on ttys005
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % mkdir homework2
bpeck@Briannas-MacBook-Pro brianna % cd ..
bpeck@Briannas-MacBook-Pro python_decal_fa25 % pwd
/Users/bpeck/python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % git clone https://github.com/pyth
ondecad/course_assignments.git
Cloning into 'course_assignments'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna          course_assignments  lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd course_assignments
bpeck@Briannas-MacBook-Pro course_assignments %
```

call ls

To see the contents, we will
call ls

You will see a folder called
homework2

DON'T mistake this folder,
for your other homework2
folder in your yourname
repository

```
course_assignments — -zsh — 80x24
Last login: Sun Jul 27 06:31:08 on ttys005
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % mkdir homework2
bpeck@Briannas-MacBook-Pro brianna % cd ..
bpeck@Briannas-MacBook-Pro python_decal_fa25 % pwd
/Users/bpeck/python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % git clone https://github.com/pyth
ondecad/course_assignments.git
Cloning into 'course_assignments'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna          course_assignments  lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd course_assignments
bpeck@Briannas-MacBook-Pro course_assignments % ls
homework2
bpeck@Briannas-MacBook-Pro course_assignments %
```

call cd

Now we will enter the homework2 folder, call cd again

```
course_assignments — -zsh — 80x24
Last login: Sun Jul 27 06:31:08 on ttys005
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % mkdir homework2
bpeck@Briannas-MacBook-Pro brianna % cd ..
bpeck@Briannas-MacBook-Pro python_decal_fa25 % pwd
/Users/bpeck/python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % git clone https://github.com/pyth
ondecad/course_assignments.git
Cloning into 'course_assignments'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna          course_assignments  lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd course_assignments
bpeck@Briannas-MacBook-Pro course_assignments % ls
homework2
bpeck@Briannas-MacBook-Pro course_assignments % cd homework2
```


call ls

To see what's inside, we will again call ls

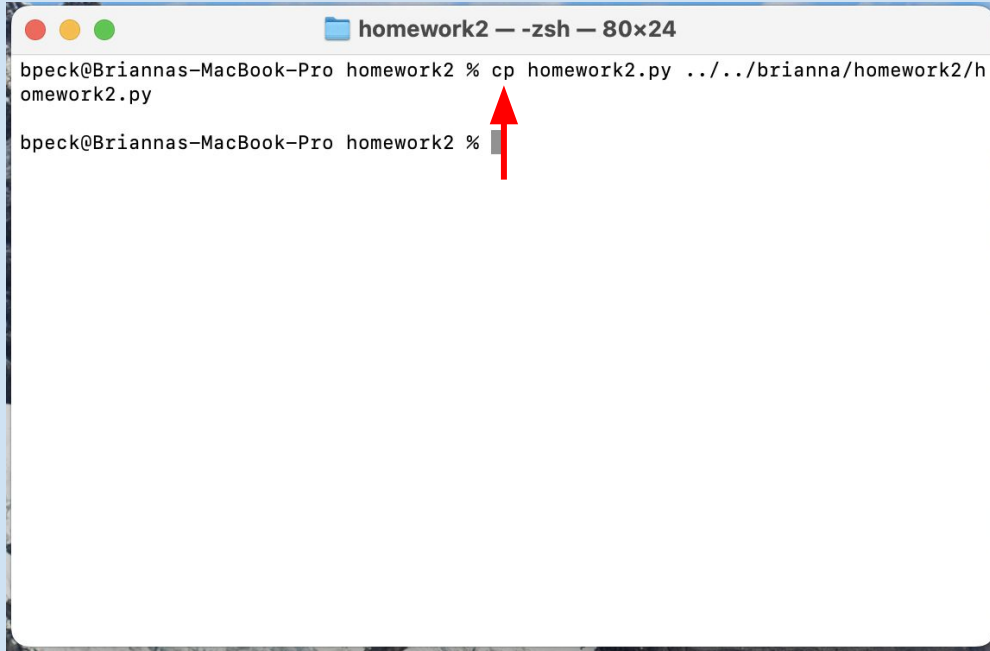
We see a file called homework2.py, our next step is to copy this file to the homework2 folder in the yourname repository

```
homework2 — -zsh — 80x24
Last login: Sun Jul 27 06:31:08 on ttys005
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % mkdir homework2
bpeck@Briannas-MacBook-Pro brianna % cd ..
bpeck@Briannas-MacBook-Pro python_decal_fa25 % pwd
/Users/bpeck/python_decal_fa25
bpeck@Briannas-MacBook-Pro python_decal_fa25 % git clone https://github.com/pyth
ondecad/course_assignments.git
Cloning into 'course_assignments'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
bpeck@Briannas-MacBook-Pro python_decal_fa25 % ls
brianna          course_assignments  lecture_notes
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd course_assignments
bpeck@Briannas-MacBook-Pro course_assignments % ls
homework2
bpeck@Briannas-MacBook-Pro course_assignments % cd homework2
bpeck@Briannas-MacBook-Pro homework2 % ls
homework2.py
bpeck@Briannas-MacBook-Pro homework2 %
```

call CP

To copy a file we will use the cp command, this is the syntax

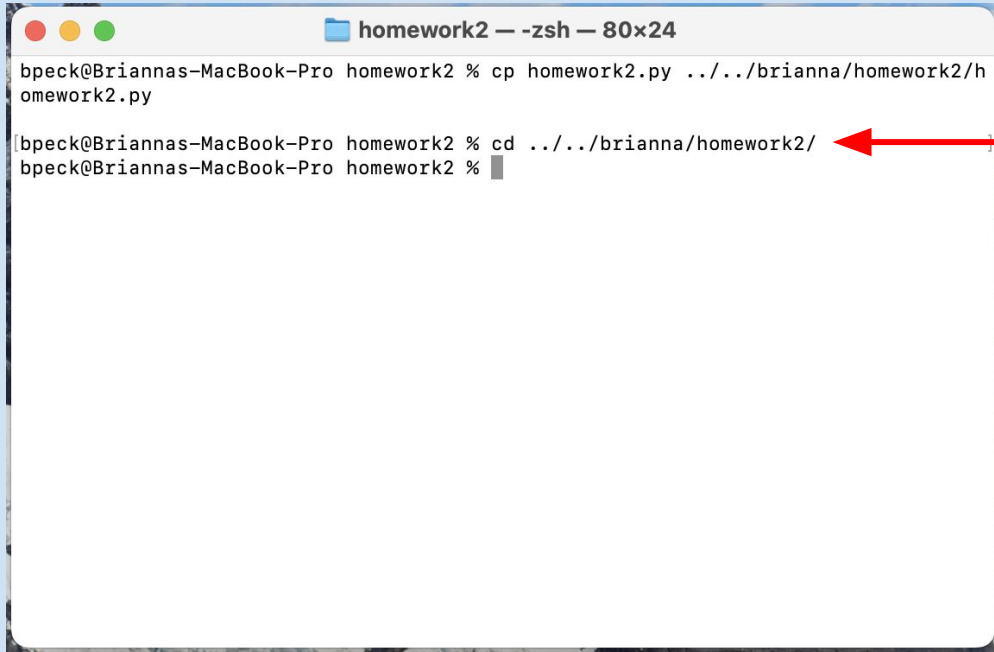
cp <filename> <new_path>

A terminal window titled "homework2 — -zsh — 80x24" is shown. The prompt is "bpeck@Briannas-MacBook-Pro homework2 %". The first command entered is "cp homework2.py ../../brianna/homework2/homework2.py", which is split across two lines. The second line shows the prompt "bpeck@Briannas-MacBook-Pro homework2 %" with a red arrow pointing to the cursor position.

```
homework2 — -zsh — 80x24
bpeck@Briannas-MacBook-Pro homework2 % cp homework2.py ../../brianna/homework2/h
omework2.py
bpeck@Briannas-MacBook-Pro homework2 %
```

call cd

To move back to yourname/homework2 we can write a relative path from where we are

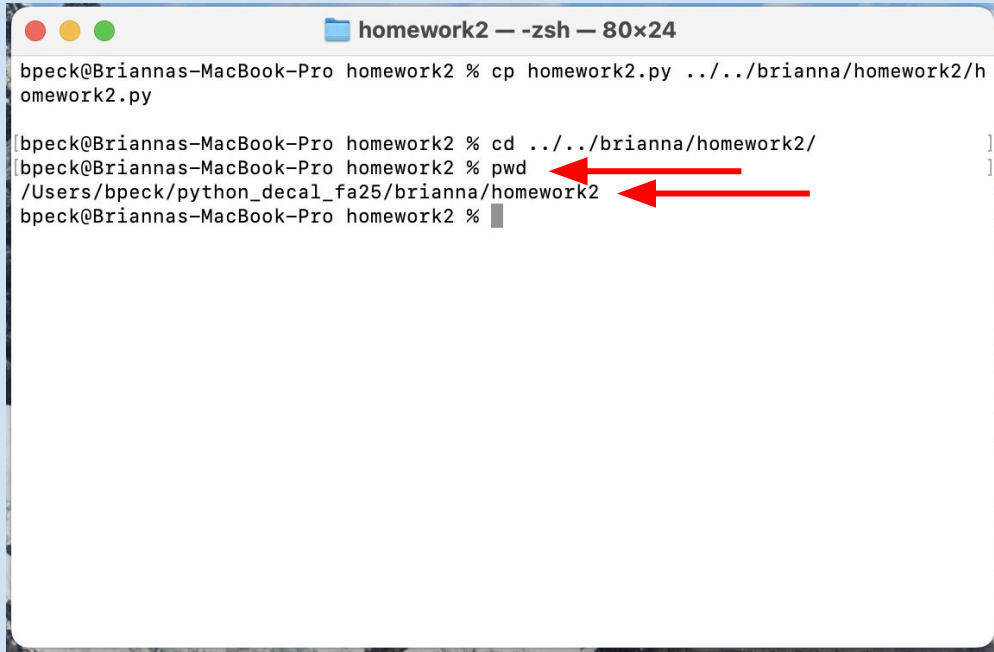


```
homework2 — -zsh — 80x24
bpeck@Briannas-MacBook-Pro homework2 % cp homework2.py ../../brianna/homework2/homework2.py
bpeck@Briannas-MacBook-Pro homework2 % cd ../../brianna/homework2/
bpeck@Briannas-MacBook-Pro homework2 %
```

A terminal window titled "homework2 — -zsh — 80x24" is shown. It contains two lines of command execution. The first line shows a file copy command: `cp homework2.py ../../brianna/homework2/homework2.py`. The second line shows a directory change command: `cd ../../brianna/homework2/`. A red arrow points from the text "can write a relative path" to the relative path `../../brianna/homework2/` in the second command.

CALL PWD

As a sanity check, we can see that we are in our **yourname/homework2** repository



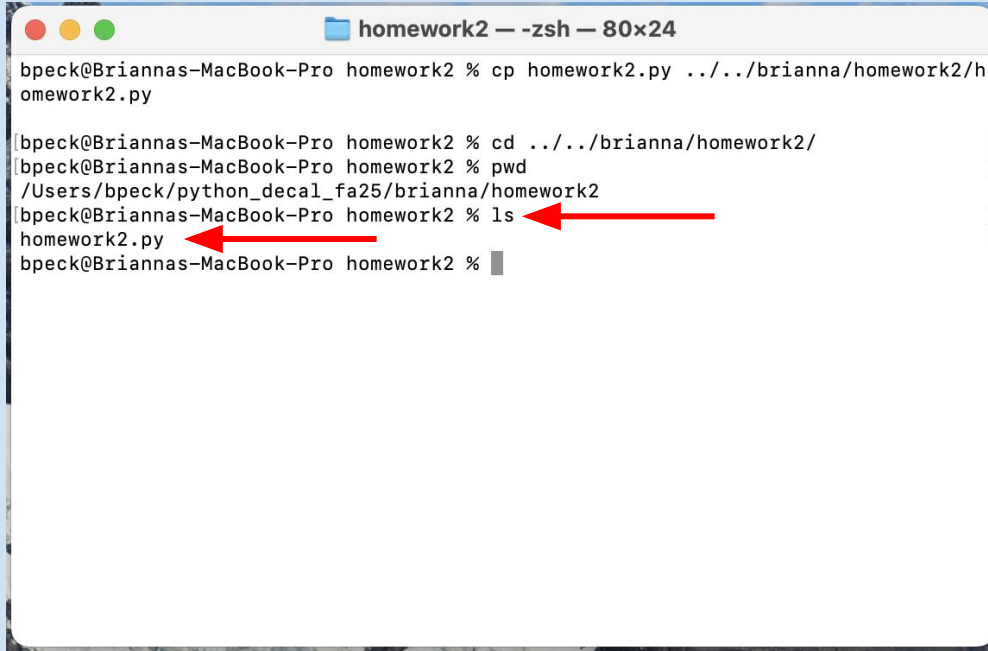
```
homework2 — -zsh — 80x24
bpeck@Briannas-MacBook-Pro homework2 % cp homework2.py ../../brianna/homework2/homework2.py
bpeck@Briannas-MacBook-Pro homework2 % cd ../../brianna/homework2/
bpeck@Briannas-MacBook-Pro homework2 % pwd
/Users/bpeck/python_decal_fa25/brianna/homework2
bpeck@Briannas-MacBook-Pro homework2 %
```

The terminal window shows a series of commands and their outputs. The first command is `cp homework2.py ../../brianna/homework2/homework2.py`. The second command is `cd ../../brianna/homework2/`. The third command is `pwd`, which outputs `/Users/bpeck/python_decal_fa25/brianna/homework2`. Two red arrows point from the text on the right to the output of the `pwd` command.

call ls

Calling ls we see the homework2.py file that we copied

The next problem asks you to open this file and answer the questions inside

A terminal window titled "homework2 — -zsh — 80x24" is shown. It contains the following commands and output:

```
bpeck@Briannas-MacBook-Pro homework2 % cp homework2.py ../../brianna/homework2/homework2.py

bpeck@Briannas-MacBook-Pro homework2 % cd ../../brianna/homework2/
bpeck@Briannas-MacBook-Pro homework2 % pwd
/Users/bpeck/python_decal_fa25/brianna/homework2
bpeck@Briannas-MacBook-Pro homework2 % ls
homework2.py
bpeck@Briannas-MacBook-Pro homework2 %
```

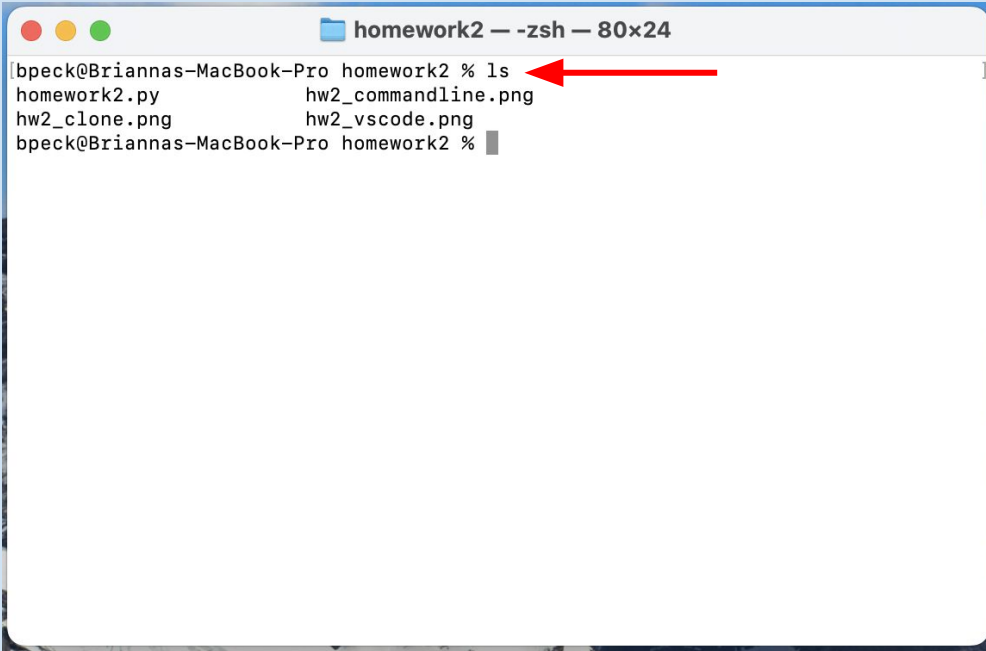
Two red arrows point to the output of the 'ls' command: one points to the filename 'homework2.py' and the other points to the 'ls' command itself.

PROBLEM #5

call ls

It's best practice to be saving your work more frequently than just after completing the entire assignment

But if this is your final change, make sure you have the following:

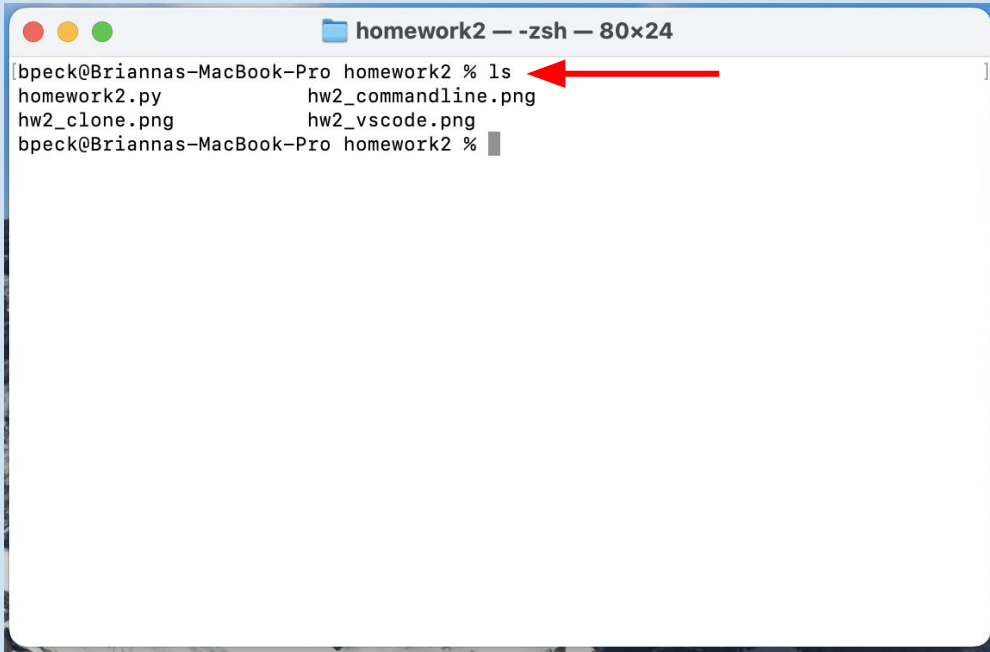
A screenshot of a terminal window titled "homework2 — zsh — 80x24". The window shows the command "ls" being executed, resulting in a two-column list of files: "homework2.py", "hw2_clone.png", "hw2_commandline.png", and "hw2_vscode.png". A red arrow points to the "ls" command in the prompt line.

```
homework2 — zsh — 80x24
bpeck@Briannas-MacBook-Pro homework2 % ls
homework2.py          hw2_commandline.png
hw2_clone.png         hw2_vscode.png
bpeck@Briannas-MacBook-Pro homework2 %
```

NOTE ABOUT SCREENSHOT

Note: After we run through how to save your changes.

You will need to take another screenshot and then save your changes again

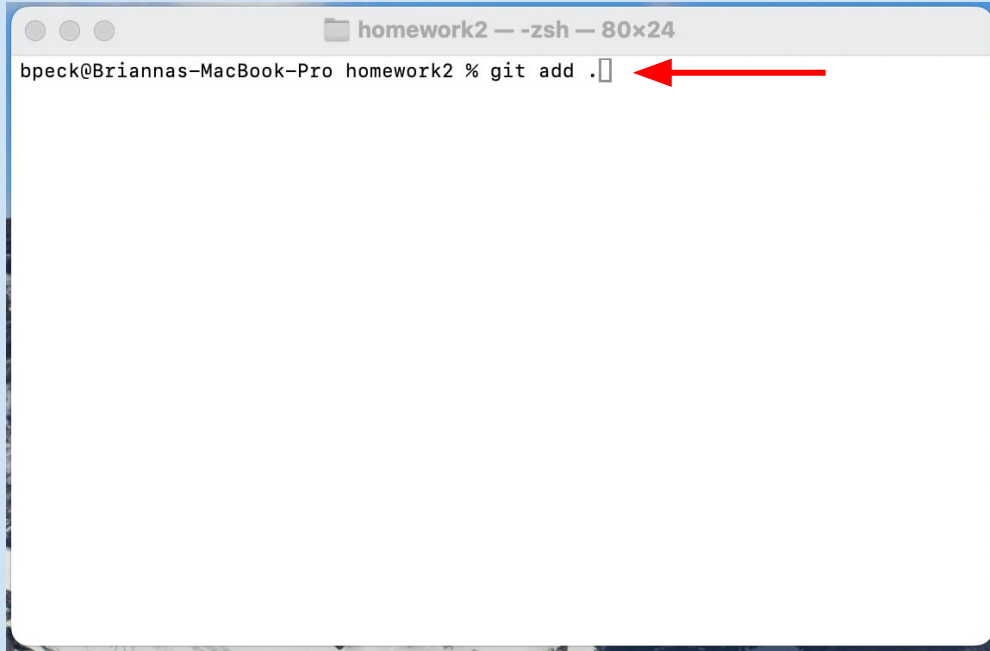
A screenshot of a terminal window titled "homework2 — zsh — 80x24". The window shows the output of the command "ls" in a directory named "homework2". The output lists four files: "homework2.py", "hw2_commandline.png", "hw2_clone.png", and "hw2_vscode.png". A red arrow points to the "hw2_commandline.png" file. The prompt "bpeck@Briannas-MacBook-Pro homework2 %" is visible at the bottom of the terminal.

```
homework2 — zsh — 80x24
bpeck@Briannas-MacBook-Pro homework2 % ls
homework2.py          hw2_commandline.png
hw2_clone.png         hw2_vscode.png
bpeck@Briannas-MacBook-Pro homework2 %
```


ADD TO STAGING Area

The first step is to call:
git add .

Because we want to add all
the files in our repository



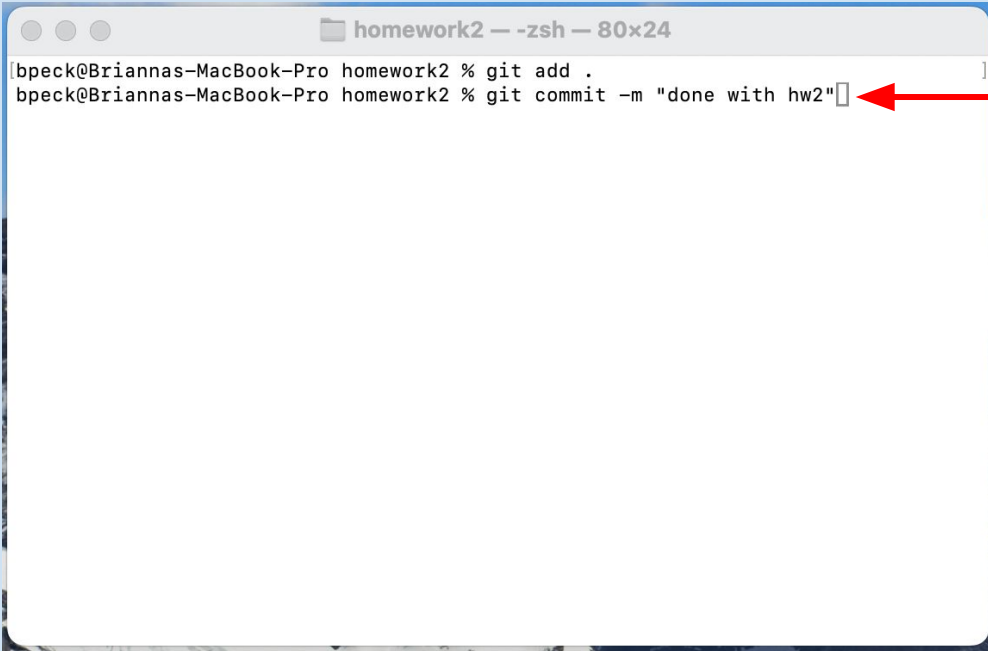
```
homework2 — zsh — 80x24
bpeck@Briannas-MacBook-Pro homework2 % git add .
```

A terminal window titled "homework2 — zsh — 80x24" is shown. The prompt is "bpeck@Briannas-MacBook-Pro homework2 %". The command "git add ." has been entered. A red arrow points to the dot at the end of the command.

commit the changes

The second step is to call:
git commit -m "message"

This takes your files from the
staged area and commits
them to your version history

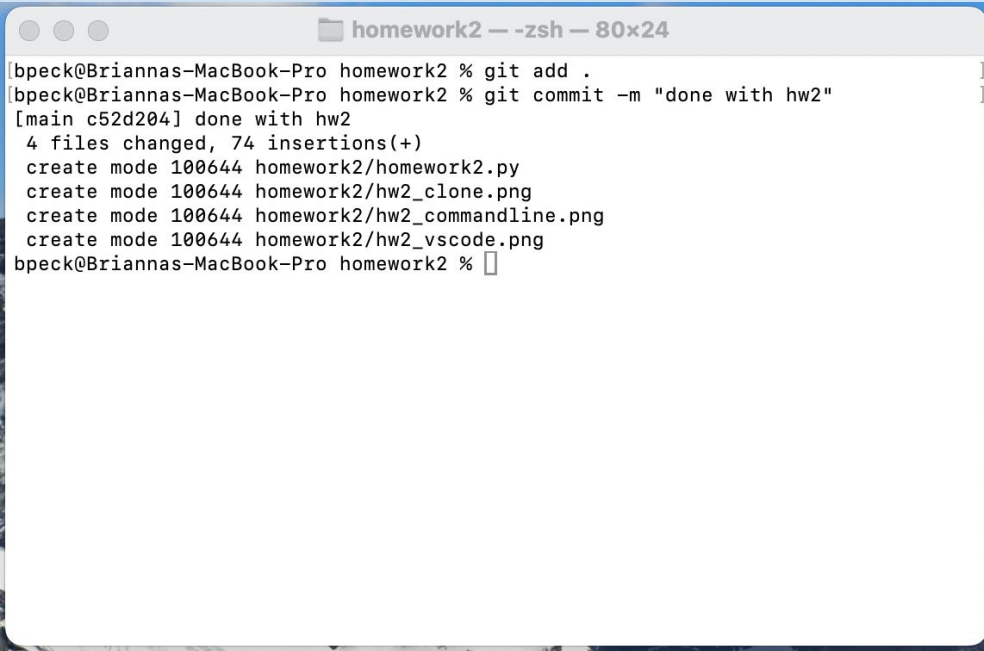
A terminal window titled 'homework2 — zsh — 80x24' showing two commands: 'git add .' and 'git commit -m "done with hw2"'. A red arrow points to the end of the second command line.

```
homework2 — zsh — 80x24
bpeck@Briannas-MacBook-Pro homework2 % git add .
bpeck@Briannas-MacBook-Pro homework2 % git commit -m "done with hw2"
```

commit the changes

Once you have committed the changes to your local repository

The next step is to push these changes to your remote repository

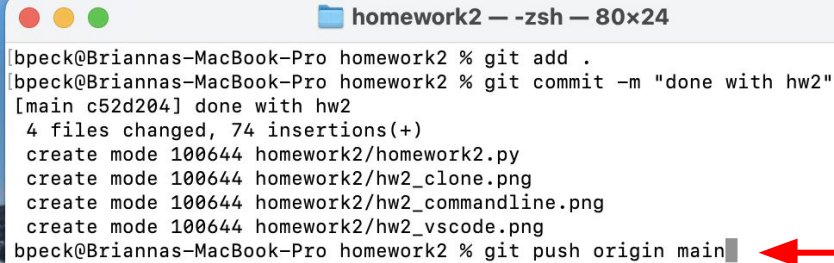
A terminal window titled 'homework2 — -zsh — 80x24' showing the execution of git commands. The user runs 'git add .' and 'git commit -m "done with hw2"'. The output shows the commit is successful, with 4 files changed and 74 insertions. The files listed are homework2.py, hw2_clone.png, hw2_commandline.png, and hw2_vscode.png.

```
homework2 — -zsh — 80x24
bpeck@Briannas-MacBook-Pro homework2 % git add .
bpeck@Briannas-MacBook-Pro homework2 % git commit -m "done with hw2"
[main c52d204] done with hw2
 4 files changed, 74 insertions(+)
 create mode 100644 homework2/homework2.py
 create mode 100644 homework2/hw2_clone.png
 create mode 100644 homework2/hw2_commandline.png
 create mode 100644 homework2/hw2_vscode.png
bpeck@Briannas-MacBook-Pro homework2 %
```

commit THE changes

Call:

git push origin main

A terminal window titled "homework2 — -zsh — 80x24" showing the execution of git commands. The output shows that 4 files were added to the repository. A red arrow points to the final command, "git push origin main".

```
bpeck@Briannas-MacBook-Pro homework2 % git add .
bpeck@Briannas-MacBook-Pro homework2 % git commit -m "done with hw2"
[main c52d204] done with hw2
4 files changed, 74 insertions(+)
create mode 100644 homework2/homework2.py
create mode 100644 homework2/hw2_clone.png
create mode 100644 homework2/hw2_commandline.png
create mode 100644 homework2/hw2_vscode.png
bpeck@Briannas-MacBook-Pro homework2 % git push origin main
```